081-06-06

06-06

PNS

Probability and Statistics (SNK)

Sudan Khatri

Demerits of primary data

↳ Biasness (researcher biasness / respondent biasness),

↳ Time, cost high

Merits of secondary

↳ low time, cost

[illegible]

Use of statistics

↳ Data analysis

↳ research making

↳ Comparison both two variables.

↳ Machine learning

↳ Drawing conclusion

↳

Imp foreram

Define, use and limitation of statistics

Define data, qualitative, quantitative nature, sources with merits and demerits.

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* Ungrouped data

x	5	10	15	20	25
f	2	3	4	2	1

* Grouped data

class	0-10	10-20	20-30	30-40	40-50
freq.	2	5	7	2	1

* Inclusive data

class	0-9	10-19	20-29	30-39	40-49
freq.	2	4	3	2	1

* Exclusive data

class	0-5	5-10	10-15	15-20	20-25
freq.	3	4	5	2	1

* Open-ended data

class	below 10	10-20	20-30	30-40	above 40
freq.	2	3	4	2	1

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* Data Series :

1) Individual series :- x :- 2, 4, 6, 8, 10, 12

2) Discrete series :-

x	5	10	15	20	25
f	2	4	2	3	1

3) Continuous series :-

class	0-10	10-20	20-30	30-40	40-50
freq	2	3	2	4	1

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Conversion of inclusive class interval into exclusive:

let, $x = \frac{2^{nd} \text{ class lower limit} - 1^{st} \text{ class upper limit}}{2}$.

lower limit = $-x$
upper limit = $+x$.

Inclusive ⇒ Exclusive

class	freq
0-9	2
10-19	1
20-29	3
30-39	2
40-49	1

0-9.5 or below 9.5 } i = 10
9.5-19.5
19.5-29.5
29.5-39.5
39.5-49.5

* Cumulative frequency:
1) less than c.f (ogive curve).
2) more than c.f (ogive curve)

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class	0-10	10-20	20-30	30-40	40-50
freq	2	4	8	5	3

1) Calculate less than c.f, more than c.f, relative frequency, % of r.f
2) Construct ogive curve and locate the median

Relative frequency = $\frac{\text{frequency of class}}{\text{sum of frequency}}$ · % of r.f = r.f × 100

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Soln.

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Class	freq.	less than c.f	More than c.f.	Relative freq.	% of rf
0-10	2	2	22	2/22	2/22 * 100%
10-20	4	6	20	4/22	4/22 * 100%
20-30	8	14	16	8/22	8/22 * 100%
30-40	5	19	8	5/22	5/22 * 100%
40-50	3	22	3	3/22	3/22 * 100%

$\Sigma f = 10$

$= 22$

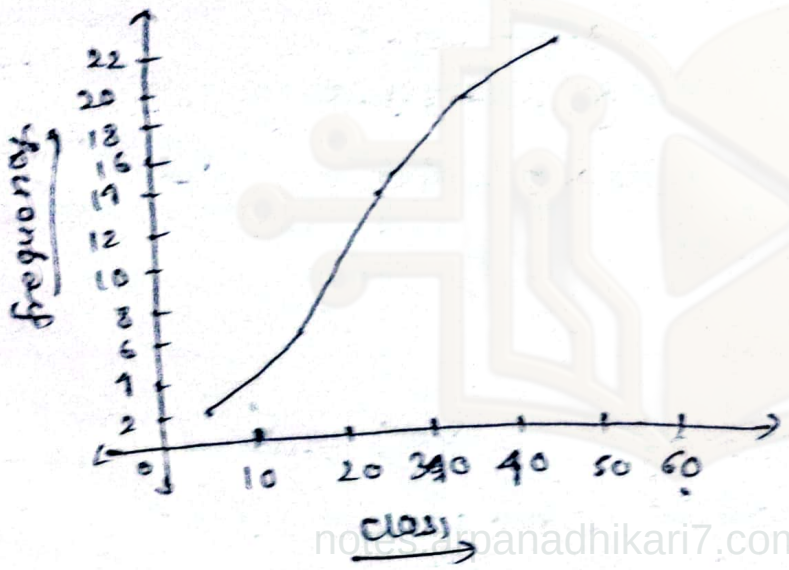

fig:- less than c.f

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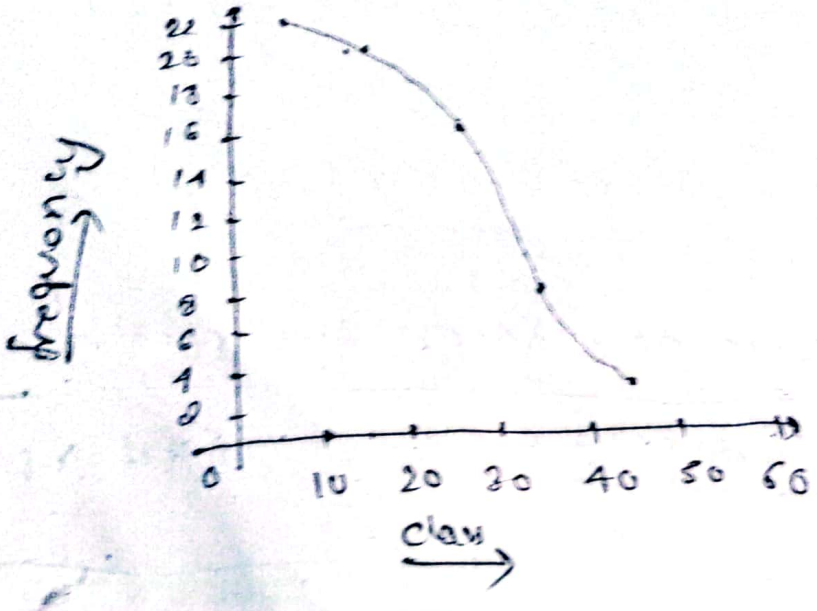

fig:- More than c.f.

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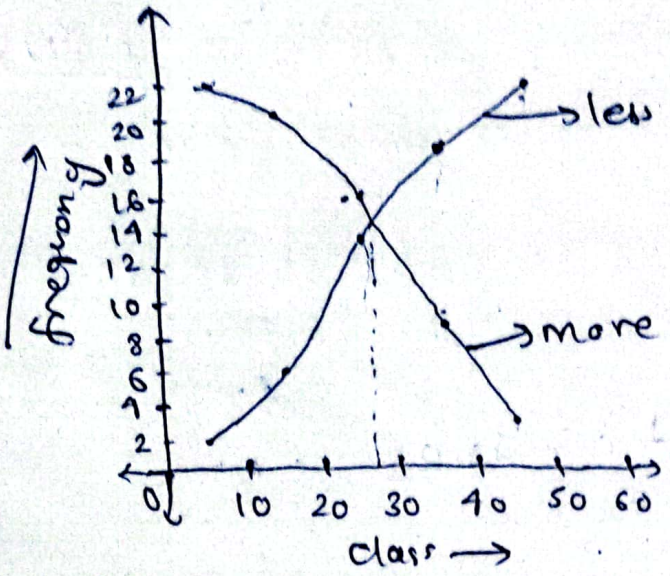

fig:- Ogive curve

Conclusion:

The median lies betⁿ interval 20-30 and is 28.

081-06-07

*Struge's formula:

$$K = 1 + 3.3221 \log N$$

$$\text{Range} = X_{max} - X_{min}$$

$$\text{class width} = \frac{R}{K}$$

Q. The weight of 40 boys in a class are as follows:

138	172	145	147	150	119	158	152
157	147	102	144	105	136	164	163
126	150	146	148	145	125	146	153
173	140	135	149	140	144	132	159
168	142	128	135	138	156	142	135

i) Construct a frequency distribution

ii) Draw less than agive and find number of boys whose weight is less than 165. And also locate the median.

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Ans Soln:

$k = 1 + 3.322 \log 40$

$k = 6.32$

Range $= 173 - 102$

$= 71$

class width $= \frac{R}{K} = \frac{71}{6.32} = 11.23$

By Sturge's formula,

Our class width is 11.23, we can take constant interval of 10 as approximate value.

class	frequency	less than cf	more than cf
100-110	1	1	40
110-120	1	2	39
120-130	3	5	38
130-140	7	12	35
140-150	14	26	28
150-160	8	34	14
160-170	4	38	6
170-180	2	40	2

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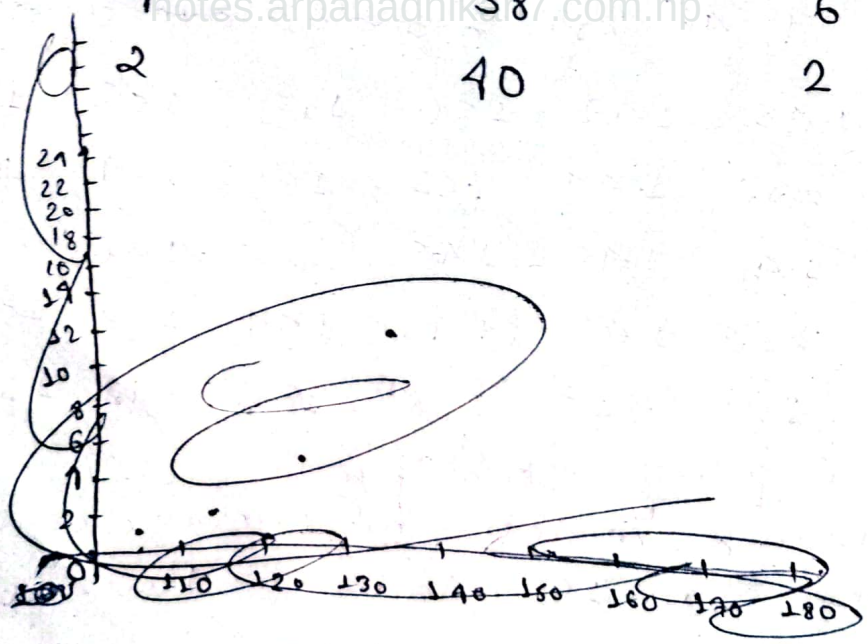

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(11)

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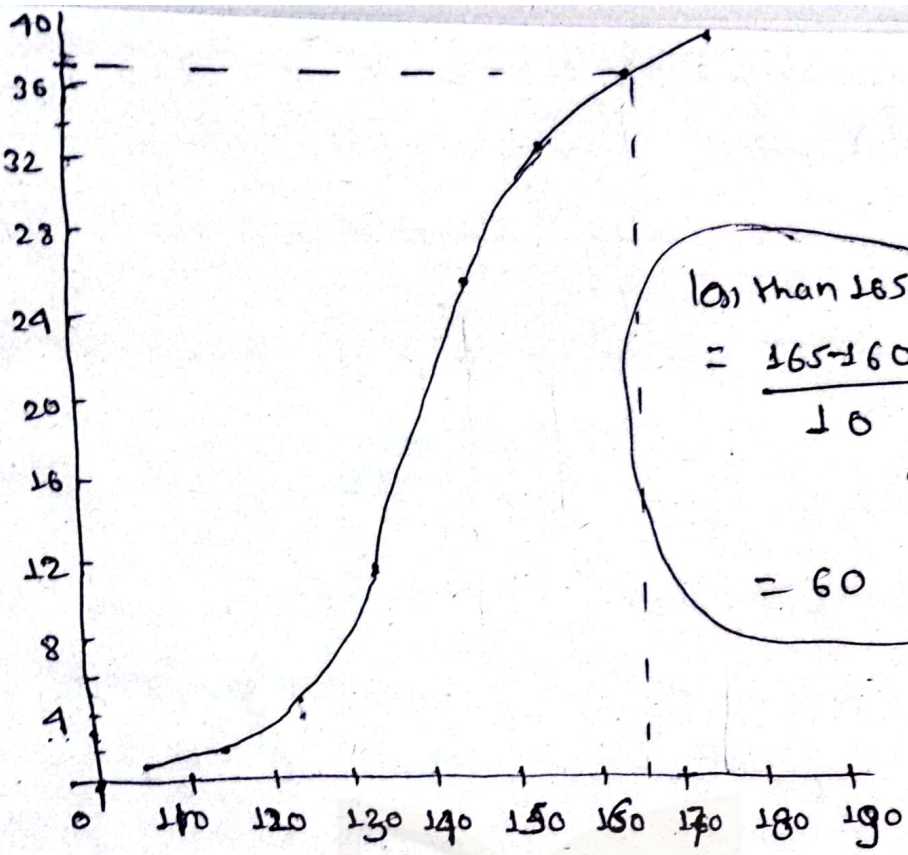
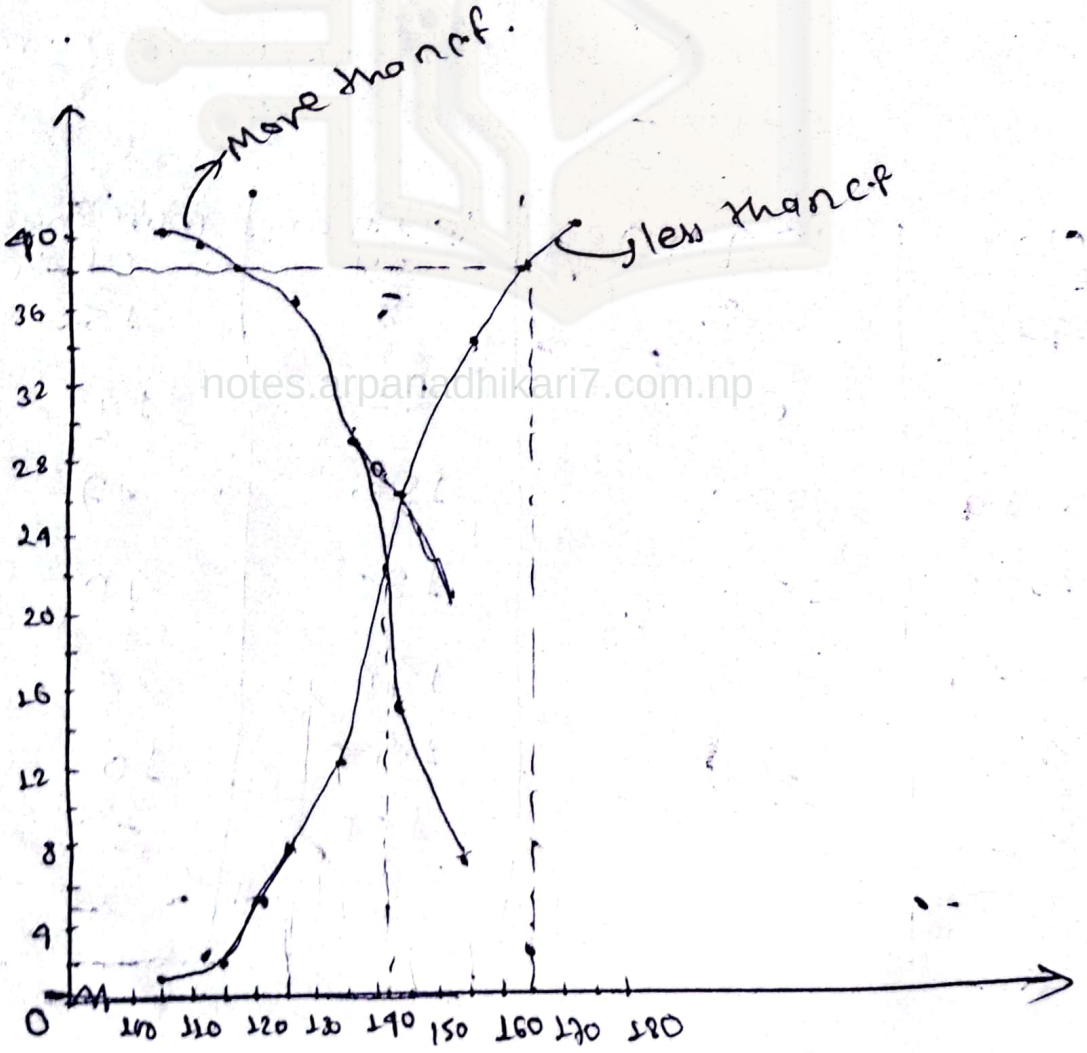


less than 165
$= \frac{165-160}{10} * 4 +$
$8+14+11+8+11+1$
$= 60$



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Q. After the implementation of an economic program to uplift the economic condition of community following information were found. 06-07 PNS

Monthly income (Rs '000')	4-6	6-8	8-10	10-12	12-14	14-16	16-18
No of families	8	65	37	15	15	5	5

Construct ogive to find

i) find number of families whose monthly income is less than 10000.

ii) find the number of families whose monthly income is between 8000 to 14000.

iii) find the number of families whose monthly income is above Rs 12000.

class	frequency	less than c.f	more than c.f
4-6	8	8	150
6-8	65	73	142
8-10	37	110	77
10-12	15	125	40
12-14	15	140	25
14-16	5	145	10
16-18	5	150	5

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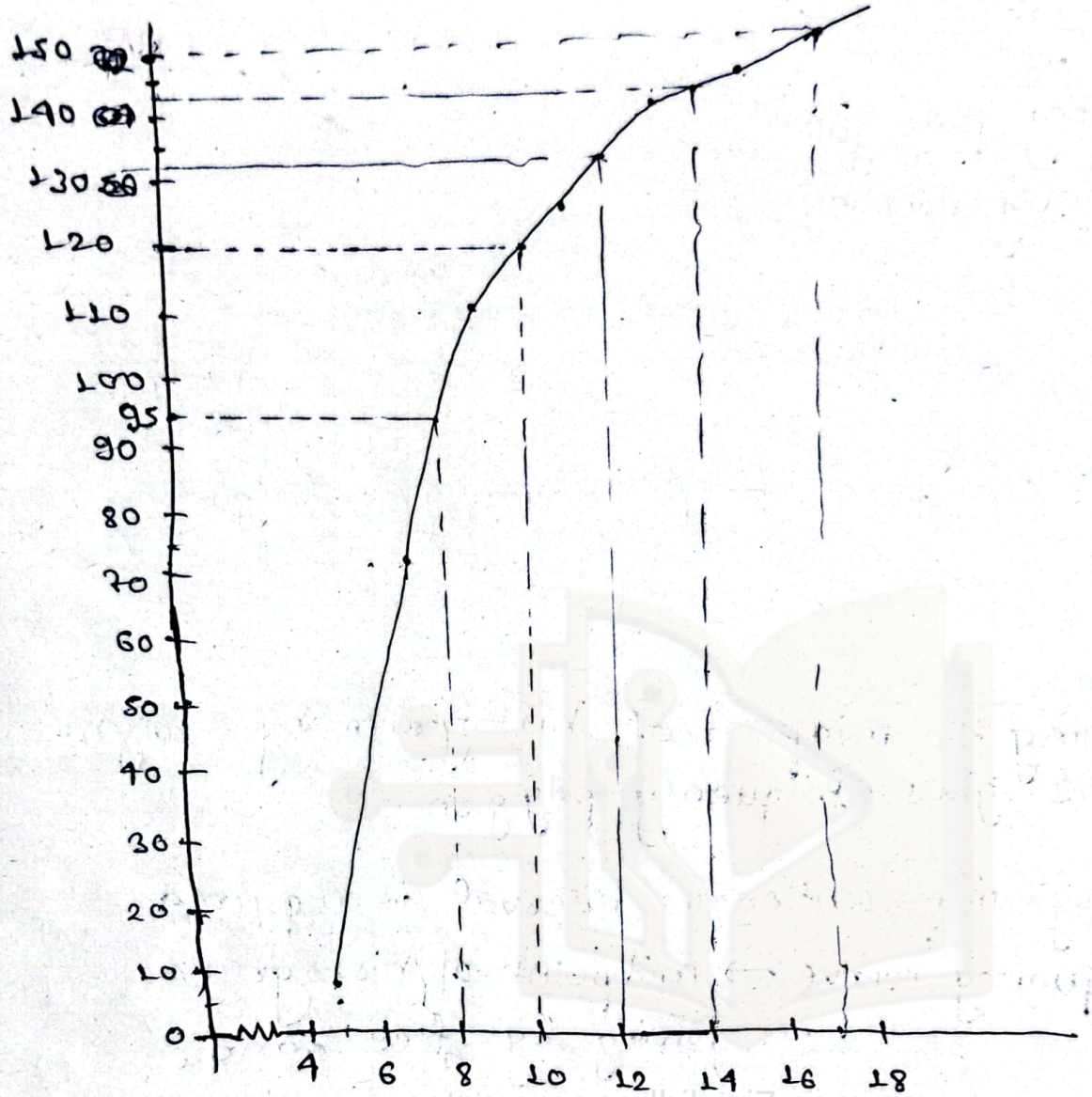

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i) From the above ogive curve, the no. of families whose monthly income is less than 10000 is 120.

ii) From the above ogive curve, the no. of families whose monthly income is between 8000 to 14000 is

$143 - 95 = 48$

iii) From the above ogive curve, the no. of families whose monthly income is above Rs 12000 is

$150 - 132 = 18$

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Histogram:

2) frequency polygon.
3) frequency curve.

Q.) class 0-10 10-20 20-30 30-40 40-50
frequency 2 5 7 3 1

i) Construct histogram, frequency polygon, frequency curve.
ii) Also locate mode.

joining the midpoint of histogram by straight line gives frequency polygon

Histogram → continuous interval is required.
frequency curve → midpoint of histogram. joining by free hand.

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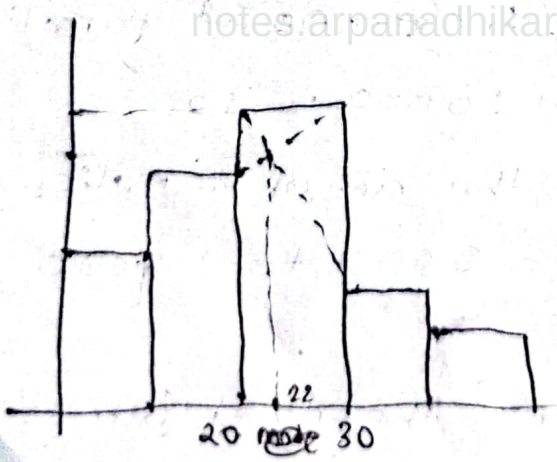

mode lies in interval 20-30 i.e. 22.

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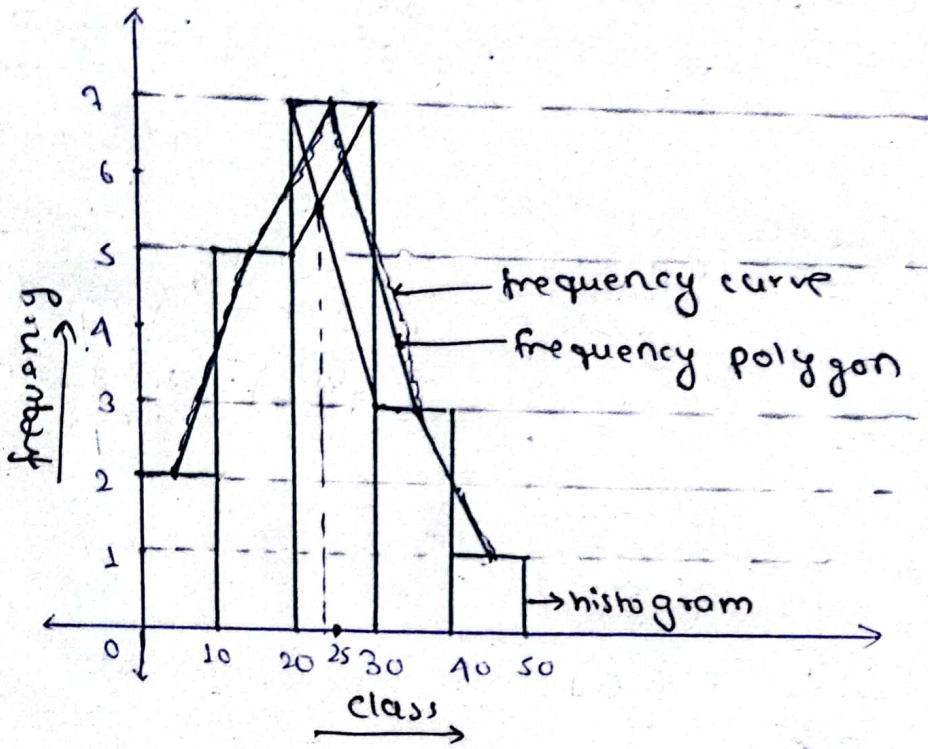
07-25 / 06-07 PAs



The mode is located in the interval 20-30 i.e. approx 24.5 according to the above figure.

081-07-25

Q. The following table shows length of eighty bally bridge.

68	84	73	82	58	90	62	88	76	93
73	79	75	73	50	93	71	59	85	75
61	65	88	87	74	52	95	78	63	72
66	78	75	75	94	77	69	74	68	60
96	78	82	67	75	95	60	79	83	71
79	62	89	97	98	85	76	65	71	75
65	80	67	57	88	78	62	76	53	74
86	67	73	81	72	63	76	75	85	77

i) Construct the group frequency distribution of following data.

ii) Construct the ogive curve and locate median.

iii) By the help of less than ogive curve, find the number of bridge having length less than 65 meters

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iv) Construct the histogram, frequency curve, frequency polygon and also locate mode. 07-25 PNS

Soln;

we have,

$$k = 1 + 3.322 \log n$$
$$= 1 + 3.322 \log 80$$
$$= 7.32$$

relative frequency = freq. of class / total sum of freq.

$$\text{Range} = 97 - 53$$
$$= 44$$

$$\text{Class width} = \frac{\ell}{k} = \frac{44}{7.32} = 6.01$$

Our class width is 6.01, we can take constant interval of 5 as approximate value.

class	frequency	less than cf.	more than cf.
50-55	1	1	80
55-60	2	3	79
60-65	11	14	77
65-70	10	24	66
70-75	12	36	56
75-80	21	57	44
80-85	6	63	23
85-90	9	72	17
90-95	4	76	8
95-100	4	80	4

$$\Sigma f = N = 80$$

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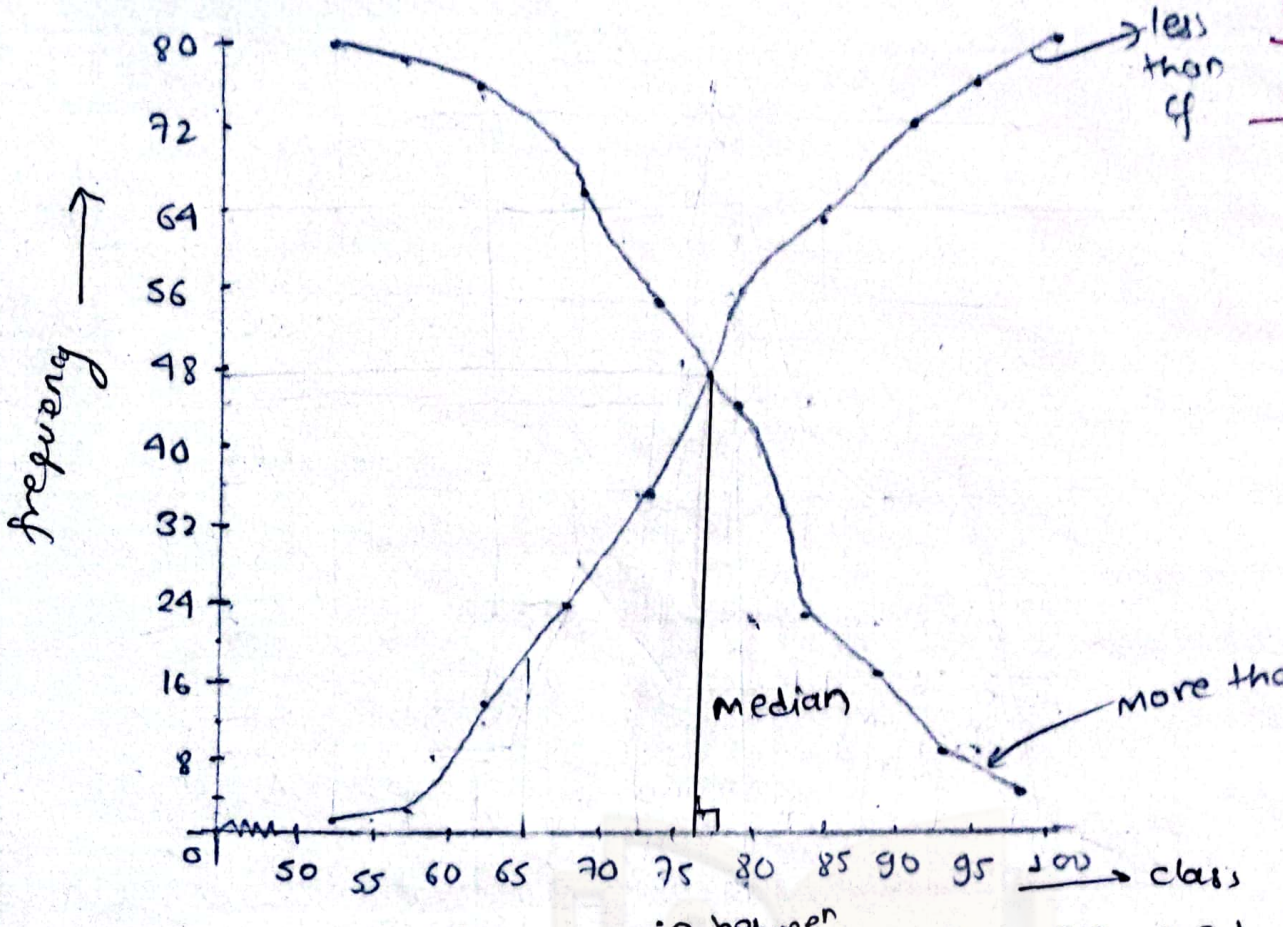
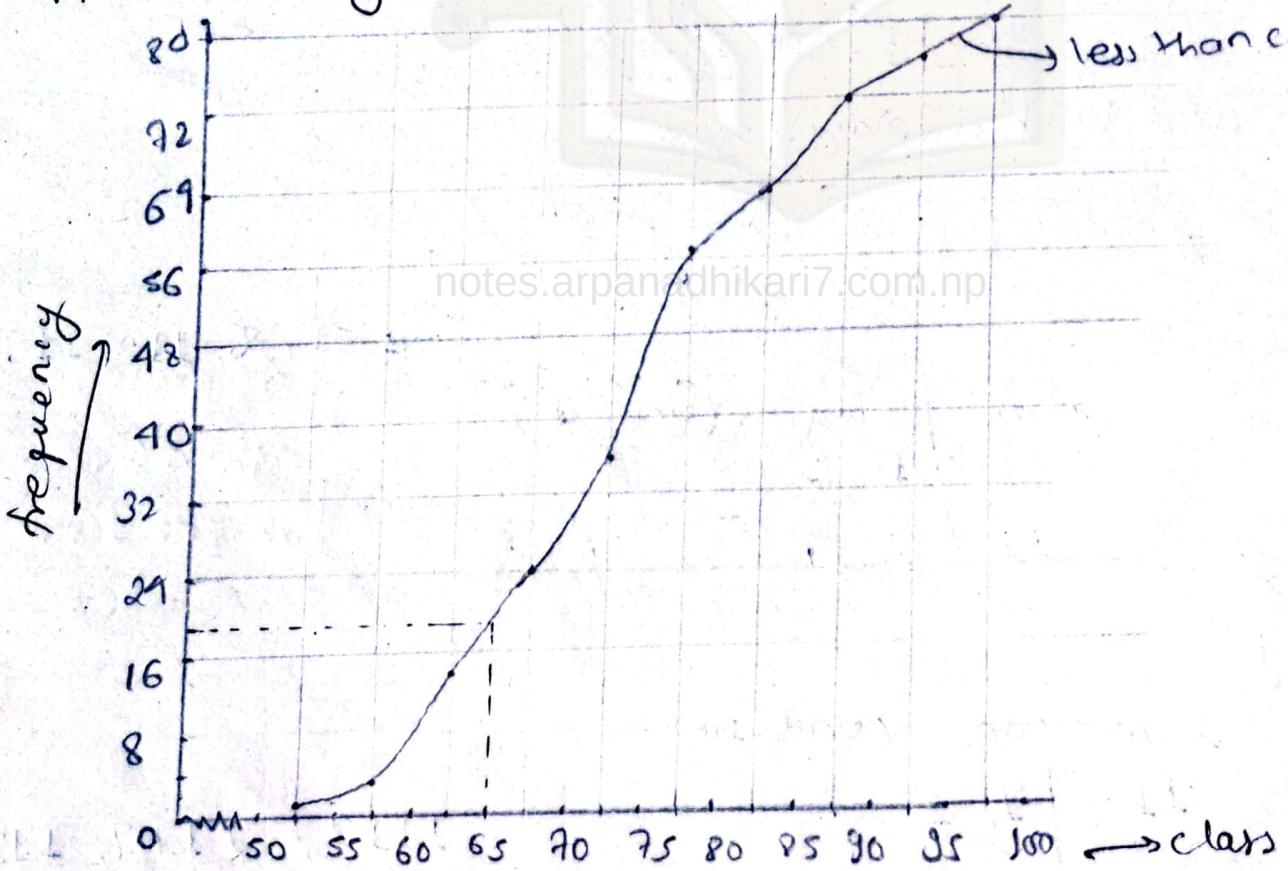


The median class lies in between class 75-80 and it is approximately 76.



By the help of less than ogive curve, the number of bridge having less than 65 meters is approx. 19.

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07-25|26
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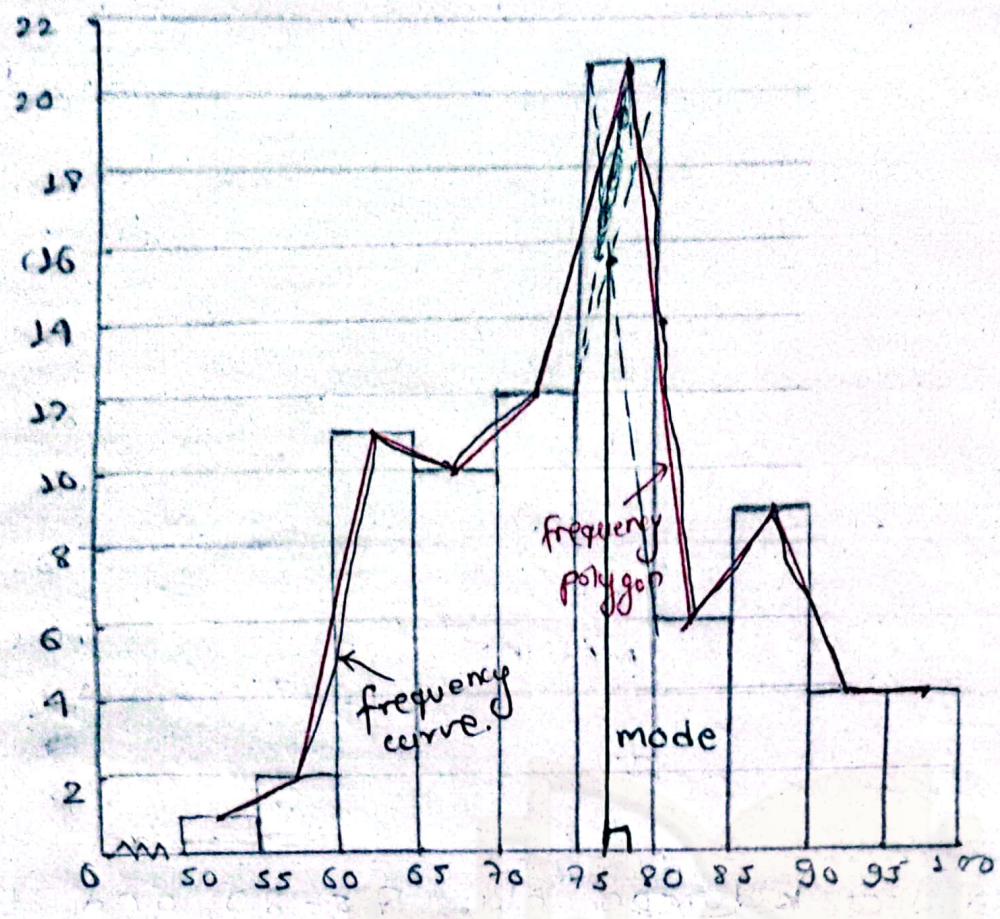

The mode is located in class interval 75-80 and it is approx 76.

081-07-26

*stem and leaf display:

~~The weights of a fruit were recorded for~~

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51, 56, 112, 114

for 2,3 digits

stem	leaf (unit 1)
5	1, 6
11	2, 4

for 4 digits

stem	leaf (unit 10)
11	1, 2

1112, 1117, 1121

Round off values → 1110, 1120, 1120

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for decimal

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stem	leaf (unit 0.1)
3	4, 5, 6

3.54, 3.56, 3.4
⇓ ⇓ ↓
3.5 3.6 3.4

Q. The weights of a fruit were recorded for a sample of 25. The data to be nearest grams are given

66 65 96 80 71 93 66 96 75 51 84
58 73 77 89 69 61 69 57 56 55 78
61 96 92

a) present the data in stem and leaf display.
b) Construct histogram.

↳ soln;
Arranging the data in ascending order

51, 55, 56, 57, 58, 61, 61, 65, 66, 66, 69, 69,
71, 73, 75, 77, 78, 80, 84, 89, 92, 93, 96, 96,
96

Now representing the order array in stem and leaf display

stem	leaf (unit 1)
5	1, 5, 6, 7, 8
6	1, 1, 5, 6, 6, 9, 9.
7	1, 3, 5, 7, 8
8	0, 4, 9.
9	2, 3, 6, 6, 6

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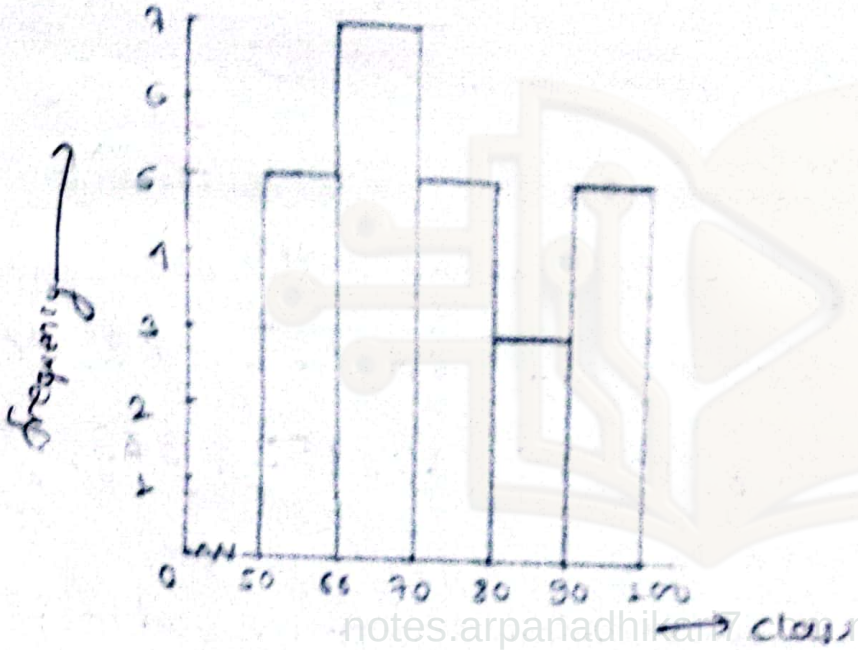
class	frequency
50-60	5
60-70	7
70-80	5
80-90	3
90-100	5

Constructing histogram:



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(9.) The following is a stem and leaf display representing the annual of gasoline purchased by gasoline for sample of 25 randomly selected service

stem	leaf
9	1, 4, 7
10	0, 2, 2, 3, 8
11	1, 2, 5, 5, 6, 6, 7, 9, 9
12	2, 2, 3, 4, 8, 9
13	0, 2

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a) Place the data into an ordered array. 07-26 PNS

b) Which of these two display seems to provide more information

c) What amount of gasoline is most likely to be purchased

d) Is there a concentration of purchase amount in centre of distribution?

↳ Soln;

a) Representing the steam and leaf display in ordered array.

91, 94, 97, 100, 102, 102, 103, 108, 111, 112, 115, 115, 116, 116, 117, 117, 117, 122, 122, 123, 124, 128, ~~[illegible]~~ 129, 130, 132

b) Among these two, ~~leaf~~ stem and leaf display seems to provide more information.

c) The amount of gasoline that is most likely to be purchased is 117.

d) ~~Median~~ we have,

$$\text{median} = \left(\frac{n+1}{2}\right)^{th} \text{item}$$

$$= \left(\frac{25+1}{2}\right)^{th} \text{item}$$

$$= 13^{th} \text{ iteam}$$

$$= 116$$

~~A) the median & mode~~

Hence, median ≠ mode

So, there is not contentration of purchase amount in centre of distribution.

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* Box and Whisker plot: Imp
* five number summary table
($x_{min}, Q_1, Md, Q_3, x_{max}$)

(a) Make a box and whisker plot of data regarding the length of a famous song

173, 206, 179, 257, 198, 251, 239, 246, 295, 181, 261

Arranging in ascending order.

173, 179, 181, 198, 206, 239, 246, 251, 257, 261, 295.

Now,

$x_{min} = 173$

$x_{max} = 295$

Median $= \left(\frac{n+1}{2}\right)^{th}$ item

$= \left(\frac{11+1}{2}\right)^{th}$ item

$= 6^{th}$ item

$= 239$

$Q_1 = \left(\frac{n+1}{4}\right)^{th}$ item

$= 3^{rd}$ item

$= 181$

$Q_3 = \left\{3\left(\frac{n+1}{4}\right)\right\}^{th}$ item

$= (3*3)^{th}$ item

$= 9^{th}$ item

$= 257$

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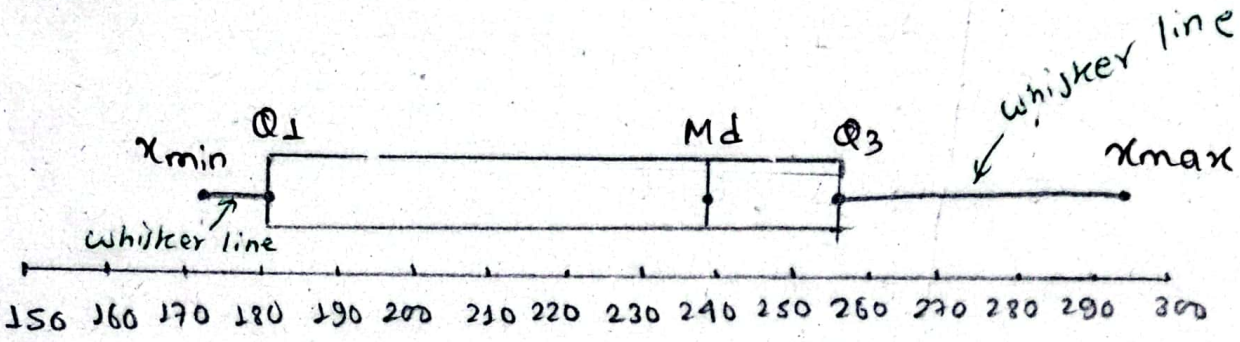
Now, Creating, five number summary table

x_{min}	Q_1	Md	Q_3	x_{max}
(173	181	239	257	295)



The distance betⁿ $(Md - x_{min})$ = 239 − 173 = 66

distance betⁿ $(x_{max} - Md)$ = 295 − 239 = 56

distance betⁿ $(Q_1 - x_{min})$ = 181 − 173 = 8

distance betⁿ $(x_{max} - Q_3)$ = 295 − 257 = 38

Here

$(Md - x_{min}) > (x_{max} - Md)$ → left skewed

$(Q_1 - x_{min}) < (x_{max} - Q_3)$ → Right skewed

Hence, the given data is not uniformly distributed.

(9) 73, 19, 16, 64, 28, 28, 31, 90, 60, 56, 22, 18, 45, 48, 17, 17, 17, 91, 92, 63, 50, 51, 31, 56, 64, 16, 17

a) Construct stem and leaf display.

b) Construct box and whisker plot and comment on nature of distribution of above data.

Soln: Arranging in ascending order

16, 16, 17, 17, 17, 17, 18, 19, 22, 28, 28, 31, 31, 45, 48, 50, 51, 56, 56, 60, 63, 64, 64, 73, 90, 91, 92

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Representing in stem and leaf display

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Stem	Leaf (unit 1)
1	6, 6, 7, 7, 7, 7, 8, 9
2	2, 8, 8
3	1, 1
4	5, 8
5	0, 1, 6, 6
6	0, 3, 4, 9
7	3
9	0, 1, 2

Here,

$x_{min} = 16$, $x_{max} = 92$

$Md = \left(\frac{n+1}{2}\right)^{th}$ item

$= \left(\frac{27+1}{2}\right)^{th}$ item

$= 14^{th}$ item

$= 45$

$Q_1 = \left(\frac{n+1}{4}\right)^{th}$ item

$= \left(\frac{27+1}{4}\right)^{th}$ item

$= 7^{th}$ item

$= 18$

$Q_3 = \left(\frac{3(n+1)}{4}\right)^{th}$ item

$= 21^{th}$ item

$= 63$

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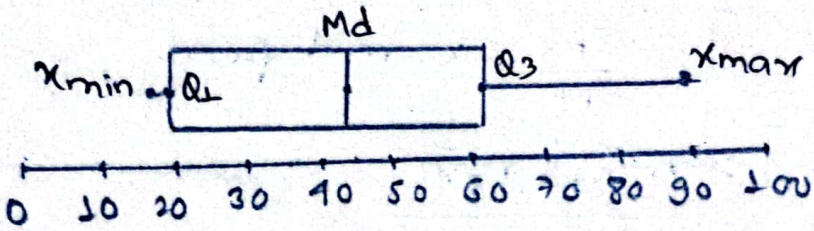
Constructing five number summary table

x_{min}	Q_1	Md	Q_3	x_{max}
16	18	45	63	92



Distance betn $(Md - x_{min}) = 45 - 16 = 29$

Distance betn $(x_{max} - Md) = 92 - 45 = 47$

Distance betn $(Q_1 - x_{min}) = 18 - 16 = 2$

Distance betn $(x_{max} - Q_3) = 92 - 63 = 29$

Here,

$(Md - x_{min}) < (x_{max} - Md)$ → Right skewed

$(Q_1 - x_{min}) < (x_{max} - Q_3)$ → Right skewed

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Hence, the given data distribution is right skewed. B

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* Measure of Central Tendency:

Mean:

1. Individual series :- $\bar{x} = \frac{\Sigma x}{n}$

2. Discrete series :- $\bar{x} = \frac{\Sigma fx}{N}$, $\Sigma f = N$

3. Continuous series :- $\bar{x} = \frac{\Sigma fx}{N}$, x = mid-point

Combined Mean $(\bar{x}_{12}) = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$

Median:

1. Individual series :- $Md = \left(\frac{n+1}{2}\right)^{th}$ item

2. Discrete series :- $Md = \left(\frac{N+1}{2}\right)^{th}$ item

3. Continuous series :- Md position $= \left(\frac{N}{2}\right)^{th}$ item

$$Md = L + \frac{\frac{N}{2} - c.f}{f} * h$$

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Mode:

1. Individual series :- Mode is the highest repeated frequency.

2. Discrete series :- Mode is the highest repeated frequency with corresponding x's variable.

3. Continuous series :- $Mo = L + \frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} * h$

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Q. Calculate mean, median, mode of following data.

1) x: 5, 7, 12, 9, 25, 4, 17, 5

↳ soln; Arranging: x: 4, 5, 5, 7, 9, 12, 17, 25

$$\text{Mean } (\bar{x}) = \frac{\Sigma x}{n} = \frac{84}{8} = 10.5$$

$$\text{Median (Md)} = \left(\frac{n+1}{2}\right)^{th} \text{ item}$$

$$= (4.5)^{th} \text{ item}$$

$$= \frac{7+9}{2} = 8$$

Mode (Mo) = 5

2)

x	5	10	15	20	25	30
f	2	4	5	7	3	1

↳ soln;

x	f	fx
5	2	10
10	4	40
15	5	75
20	7	140
25	3	75
30	1	30
	Σf = 22	Σfx = 370

$$\bar{x} = \frac{\Sigma fx}{\Sigma f} = \frac{370}{22} = 16.818$$

$$Md = \left(\frac{N+1}{2}\right)^{th} \text{ item} = \left(\frac{22+1}{2}\right)^{th} \text{ item} = (11.5)^{th} \text{ item} = \frac{15+20}{2} = 17.5$$

The highest frequency is 7 and its corresponding x value is 20

So, Mo = 20

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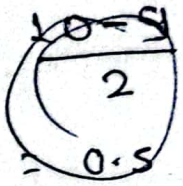
3.) class 0-9 10-19 20-29 30-39 40-19 07-27 PNS

No of students 3 5 9 7 2

class	No. of students
0-9	3
10-19	5
20-29	9
30-39	7
40-49	2
	N = 26



Converting into exclusive interval.

$$\frac{10-9}{2} = 0.5$$

class	No. of students (f)	x	fx	cf
0-9.5	3	4.75	14.25	3
9.5-19.5	5	14.5	72.5	8
19.5-29.5	9	24.5	220.5	17
29.5-39.5	7	34.5	241.5	24
39.5-49.5	2	44.5	89	26
	N = 26		Σfx = 637.75	

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$$\bar{x} = \frac{637.75}{26} = 24.5288$$

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07-27
PNS

Md position = $\left(\frac{N}{2}\right)^{th}$ item

= 13^{th} item

The median lies in the class interval 19.5 - 29.5

so,

$L = 19.5$, $f = 9$, $h = 10$, $cf = 8$.

$$Md = L + \frac{\frac{N}{2} - cf}{f} * h$$

$$= 19.5 + \frac{13 - 8}{9} * 10$$

$$= 25.055$$

For Mode

$f_m = 9$

$f_1 = 5$

$f_2 = 7$.

The highest repeated frequency is 9. so, the mode lies in class interval 19.5 - 29.5.

so, $L = 19.5$, $h = 10$

Now,

$$M_o = L + \frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} * h$$

$$= 19.5 + \frac{9 - 5}{9 - 5 + 9 - 7} * 10$$

$$= 26.166$$ Ans

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Q. The mean weight of 150 students is 60 kg. The mean weight of boys is 70 kg and girls is 55 kg. find the no of boys and girls. 07-27 PNS

Soln:

Mean wt. of boys ($\bar{x}_1$) = 70

Mean wt. of boys ($\bar{x}_2$) = 55 kg

No. of boys (n_1) = ?

No. of girls (n_2) = ?

Combined mean wt. of students ($\bar{x}_{12}$) = 60.

we have,

$n_{12} = 2$

Mean wt. of 150 students ($\bar{x}_{12}$) = 60

we have,

$$\bar{x}_{12} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

$$\text{or, } 60 = \frac{n_2 \cdot 70 + 55(150 - n_1)}{150}$$

$$\therefore 9000 = 70n_1 + 8250 - 55n_1$$

$$\therefore 750 = 15n_1$$

$$\therefore n_1 = 50$$

Now,

$$n_2 = 150 - n_1$$
$$= 150 - 50$$
$$= 100$$

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(8) The mean marks of 300 students is 45. The mean marks of top 100 students is 70. The mean marks of last 100 students is 20. Find the mean marks of middle students. 07-29 PNS

by we have,

mean marks of top 100 students, $(\bar{x}_1) = 70$

mean " " last " " $(\bar{x}_2) = 20$

mean marks of middle students $(\bar{x}_3) = ?$

$n_1 = 100$

$n_2 = 100$

$n_3 = 300 - 100 - 100$

$n_3 = 100$

mean marks of 300 students $(\bar{x}_{123}) = 45$

$N = 300$

we have

$$\bar{x}_{123} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + n_3\bar{x}_3}{n_1 + n_2 + n_3}$$

$$45 = \frac{100*70 + 100*20 + 100*\bar{x}_3}{300}$$

$$13500 = 7000 + 2000 + 100\bar{x}_3$$

$$4500 = 100\bar{x}_3$$

$$\bar{x}_3 = 45$$

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07-27
PNS

(Q.) Mean of 100 items was 50. Later on, it was found that two items were misread as 60 and 8 instead of 192 and 66. Find the correct mean

Soln;

$$\text{Mean}(\bar{x}) = 50$$

$$\frac{\Sigma x}{n} = 50$$

$$\frac{\Sigma x}{100} = 50$$

$$\Sigma x = 5000$$

Case-II

$$\Sigma x' = \Sigma x - 60 - 8 + 192 + 66 = 5190$$

Now,

$$\bar{x}' = \frac{\Sigma x'}{n}$$

$$= \frac{5190}{100}$$

$$= 51.9$$

Ans

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(5) Arithmetic mean of 150 item is 50. Two item 40 and 60 were left out at the time of calculation. What is the correct mean of all item?

07-29
PNS

Soln: Case-I

Total items (n) = 150

mean, ($\bar{x}$) = 50

we have

$$\bar{x} = \frac{\Sigma x}{n}$$

$$50 = \frac{\Sigma x}{150}$$

$$\Sigma x = 7500$$

Case-II: Two items were left out.

$$\Sigma x' = \Sigma x + 40 + 60 = 7600$$

$$n' = 150 + 2 = 152$$

The correct mean

$$\bar{x}' = \frac{\Sigma x'}{n'}$$

$$= \frac{7600}{152}$$

$$= 50$$

Ans

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07-27
PNS

(Q1) The expenditure of 1000 families is given.

imp

Expenditure	40-59	60-79	80-99	100-119	120-139
No: of family	50	-	500	-	50

The median for distribution is 87. Calculate the missing frequency.

(Q.2) The marks of 100 students is given. If mode value is 58, find the missing frequency.

imp

Marks	0-20	20-40	40-60	60-80	80-100
No. of students	14	-	27	-	15

(Q.1) soln;

Converting into exclusive interval;

$\frac{60-59}{2} = 0.5$

we have,
$N = 1000$.

Expenditure	No. of family (f)	c.f
39.5 - 59.5	50	50
59.5 - 79.5	x (let)	50+x
79.5 - 99.5	500	550+x
99.5 - 119.5	y (let)	550+x+y
119.5 - 139.5	50	600+x+y.
	$N = \Sigma f = 600+x+y$	

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Median = 87.

It lies in the interval 79.5 - 99.5

So, $L = 79.5$, $f = 500$, $h = 20$, $c.f = 50+x$

Now, we have,

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PNS

$$\text{Median} = L + \frac{\frac{N}{2} - cf}{f} * h$$

or, $87 = 79.5 + \left\{ \frac{\frac{1000}{2} - (50+x)}{500} \right\} * 20$

or, $7.5 = \frac{500 - 50 - x}{500} * 20$

or $\frac{7.5 * 500}{20} = 450 - x$

or, $187.5 = 450 - x$

$\therefore x = 262.5 \approx 263$

Also, we have,

$$600 + x + y = 1000$$

or $600 + 263 + y = 1000$

$\therefore y = 137$

∴ The missing frequencies are 263 and 137. Ans

(Q.2) Soln;

Marks	No. of students (f)
0-20	14
20-40	- x (let)
40-60	27
60-80	- y (let)
80-100	15
	$\Sigma f = 56 + x + y$

we have,

$N = 100$.

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07-27
PNS

Given

Mode = 58.

It lies in class interval 40-60. So,

$L = 40,\ h = 20,\ f_m = 27,\ f_1 = x,\ f_2 = y$

We have,

$$\text{Mode} = L + \frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} \times h$$

or, $$58 = 40 + \frac{27 - x}{27 - x + 27 - y} \times 20$$

or, $$\frac{18}{20} = \frac{27 - x}{54 - x - y}$$

or, $$0.9 = \frac{27 - x}{54 - x - y}$$

or, $$48.6 - 0.9x - 0.9y = 27 - x$$

or, $$21.6 - 0.9y + 0.1x = 0$$

or, $$0.9y - 0.1x = 21.6 \quad \text{—} (1)$$

Also,

$$56 + x + y = 100$$

or, $$y = 44 - x \quad \text{—} (2)$$

from (1) and (2),

$$0.9(44 - x) - 0.1x = 21.6$$

or, $$39.6 - 0.9x - 0.1x = 21.6$$

$\therefore x = 18$

Now, $y = 44 - 18 = 26$ ✓

∴ Missing frequencies are 18 & 26. Ans

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Quartiles - Q_i

1. Individual series, $Q_i = \frac{i(n+1)}{4}^{th}$ item ; $i = 1, 2, 3$.

2. Discrete series, $Q_i = \frac{i(N+1)}{4}^{th}$ item ; $\Sigma f = N$.

3. Continuous series, Q_i position $= \frac{i(N)}{4}^{th}$ item

$$Q_i = L + \frac{\frac{iN}{4} - c.f}{f} * h$$

Deciler - D_j

1. Individual series, $D_j = \frac{j(n+1)}{10}^{th}$ item ; $j = 1, 2, 3 \dots 9$

2. Discrete series, $D_j = j\frac{(N+1)}{10}^{th}$ item

3. Continuous series, D_j position $= \frac{j(N)}{10}^{th}$ item

$$D_j = L + \frac{\frac{jN}{10} - c.f}{f} * h$$

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Percentiles - P_k

1. Individual series, $P_k = \frac{k(n+1)}{100}^{th}$ item ; $k = 1, 2, \dots, 99$

2. Discrete series, $P_k = \frac{k(N+1)}{100}^{th}$ item

3. Continuous series, P_k position $= \frac{k(N)}{100}^{th}$ item

$$P_k = L + \frac{\frac{k(N)}{100} - c.f}{f} * h$$

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Q. Calculate $Q_1, Q_3, D_3, D_7, P_{35}, P_{65}$

OR Q_1 PNS

1) x : 23, 17, 15, 25, 6, 9, 8.

2)

class (cls)	0-9	10-19	20-29	30-39	40-49
No. of students	3	8	15	13	22

① → soln; Arranging: 6, 8, 9, 15, 17, 23, 25.

$n = 7$

$Q_1 = \frac{1(7+1)}{4}$ th item

$= 2^{th}$ item

$= 8$

$Q_3 = \frac{3(7+1)}{4}$ th item

$= 6^{th}$ item

$= 23$

$D_3 = \frac{3(7+1)}{10}$ th item

$= 2.4^{th}$ item

$= 2^{nd} + 0.4(3^{rd} - 2^{nd})$

$= 8 + 0.4(9 - 8)$

$= 8.4$

$D_7 = \frac{7(7+1)}{10}$ th item

$= 5.6^{th}$ item

$= 17 + 0.6(23 - 17)$

$= 20.6$

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$$P_{35} = 35\frac{(7+1)}{100}^{th} \text{ item}$$

$$= 2.8^{th} \text{ item}$$

$$= 8 + 0.8(9-8)$$

$$= 8.8$$

$$P_{65} = 65\frac{(7+1)}{100}^{th} \text{ item}$$

$$= 5.2^{th} \text{ item}$$

$$= 17 + 0.2(23-17)$$

$$= 18.2$$

(2) → soln;

converting into exclusive:

$$\frac{10-9}{2} = 0.5$$

class (less)	No. of students (c.f)	frequency (f)
0 - 9.5	3	3
9.5 - 19.5	8	5
19.5 - 29.5	15	7
29.5 - 39.5	19	4
39.5 - 49.5	22	3
		$\Sigma f = N = 22$

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Now,

$$Q_1 \text{ position} = \left(\frac{22}{4}\right)^{th} \text{ item}$$

$$= 5.5^{th} \text{ item}$$

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It lies in interval 9.5 - 19.5. so,

$L = 9.5,\ c.f = 3,\ h = 10,\ f = 5$

$$Q_1 = 9.5 + \frac{1.(5^{2}.5 - 3)}{5} * 10$$

$$= 14.5$$

Again.

$$Q_3 \text{ position} = 3\left(\frac{N}{4}\right)^{th} \text{ item}$$

$$= 3.\left(\frac{22}{4}\right)^{th} \text{ item}$$

$$= 16.5^{th} \text{ item.}$$

It lies in class interval 29.5 - 39.5 so,

$L = 29.5,\ h = 10,\ f = 4,\ c.f = 15$

$$Q_3 = 29.5 + \frac{16.5 - 15}{4} * 10$$

$$= 33.25$$

Now,

$$D_3 \text{ position} = 3\left(\frac{22}{10}\right)^{th} \text{ item}$$

$$= 6.6^{th} \text{ item}$$

It lies in class interval. 9.5 - 19.5 . so,

$L = 9.5,\ c.f = 3,\ h = 10,\ f = 5$

$$D_3 = 9.5 + \frac{6.6 - 3}{5} * 10$$

$$= 16.7$$

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D_7 position $= 7\left(\frac{22}{10}\right)^{th}$ item

$= 15.4^{th}$ item

08-02
PNS

It lies in class interval 29.5 - 39.5 - 80,

$L = 29.5$, $h = 10$, $f = 4$, $c.f = 15$

$$D_7 = 29.5 + \frac{15.4 - 15}{4} * 10$$

$$= 30.5$$

Now

P_{35} position $= 35\left(\frac{22}{100}\right)^{th}$ item

$= 7.7^{th}$ item

It lies in interval 9.5 - 19.5 . 50,

$L = 9.5$, $c.f = 3$, $h = 10$, $f = 5$

$$P_{35} = 9.5 + \frac{7.7 - 3}{5} * 10$$

$$= 18.9$$

Again

P_{65} position $= 65\left(\frac{22}{100}\right)^{th}$ item

$= 14.3^{th}$ item

It lies in interval 19.5 - 29.5 . 50,

$L = 19.5$, $h = 10$, $f = 7$, $c.f = 8$

$$P_{65} = 19.5 + \frac{14.3 - 8}{7} * 10$$

$= 28.5$

Ans

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Q.1 120 students appeared for a certain test and following marks distribution was obtained. 08-02 PNS

IMP

Marks	0-20	20-40	40-60	60-80	80-100
Students	10	30	36	30	14

find:

a) The limits of marks of middle 30% students.

b) The percentage of students getting more than 75.

c) The number of students who fail, if 35 marks are required for passing. 48-84

Imp

Q.2 Find minimum pass marks if only 20% of students had failed and also minimum marks obtained by ~~the~~ top 25% of students.

Marks	0-20	20-40	40-50	50-60	60-80	80-100
students	50	100	150	90	60	50

a) highest marks obtained by lowest 10% → P_{10}

b) lowest mark obtained by highest 10% → P_{90}.

Q.1. ~~Solution~~

Marks	students (f)	c.f.
0-20	10	10
20-40	30	40
40-60	36	76
60-80	30	106
80-100	14	120
	$\Sigma f = N = 120$	

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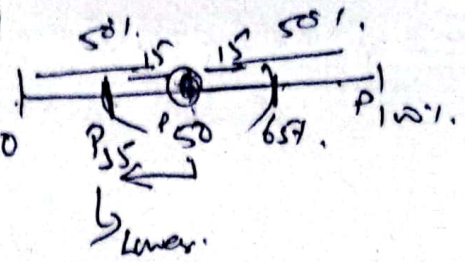
PNS

(a)



Lower.

(P_{35}, P_{65})

$\frac{30}{2} = 15\%$.

$50 - 15\% = 35\%$.

$50 + 15\% = 65\%$.

(b)

$\frac{80-75}{20} \times 30 + 14$

$7.5 + 14$

21.5

$\frac{21.5}{120} \times 100\%$.

$\frac{35-20}{30} \times \frac{1.5}{30} + 10$

$= 32.5$

~ 33

(c)

(a) The limits of marks of middle 50% students can be calculated by calculating P_{35} and P_{65}.

P_{35} position $= 35\left(\frac{120}{100}\right)^{th}$ item

$= 42^{th}$ item.

It lies in the interval 40-60. So,

$L = 40,\ h = 20,\ f = 36,\ c.f = 40$

Now,

$P_{35} = 40 + \frac{42-40}{36} * 20$

$= 41.11$

P_{65} position $= 65\left(\frac{120}{100}\right)^{th}$ item

$= 78^{th}$ item

It lies in the interval. 60-80. So.

$L = 60,\ h = 20,\ f = 30,\ c.f = 76$

$P_{65} = 60 + \frac{78-76}{30} * 20$

$= 61.33$

∴ The limits are 41.11 and 61.33 Ans

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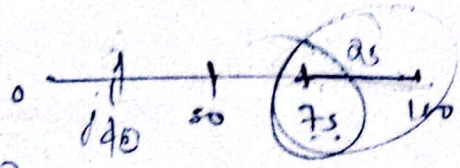
Q.2 →soln;

08-02/03
PNS

Marks	Students (f)	cf.
0-20	50	50
20-40	100	150
40-50	150	300
50-60	90	390
60-80	60	480
80-100	50	500
	$\Sigma f = N = 500$	

03%



$P_{75} = 75 \cdot \left(\frac{500}{100}\right)^{th}$

375^{th}

$50 + \frac{375 - 300}{90} \times 10$

58.33

$P_{20} = 20 \left(\frac{500}{100}\right)^{th}$

$= 100^{th}$

$20 + \frac{100 - 50}{100} \times 20$

$= 30$

081-08-03

* Measure of dispersion:

* Standard Deviation:

1) Individual series:

$$\sigma = \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2}$$

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2) Discrete series:

$$\sigma = \sqrt{\frac{\Sigma f x^2}{N} - \left(\frac{\Sigma f x}{N}\right)^2}$$

3) Continuous series:

* Variance:

$$\sigma^2$$

* Co-eff of variation (C.V) $= \frac{\sigma}{\bar{x}} * 100\%$

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PNS

* Combined standard deviation :

$$\sigma_{12} = \sqrt{\frac{n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)}{n_1 + n_2}}$$

$$d_1 = \bar{x}_1 - \bar{x}_{12}$$

$$d_2 = \bar{x}_2 - \bar{x}_{12}$$

$$\bar{x}_{12} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

Q. Following is the weights of persons.

120, 180, 140, 150, 160

Find:

a) standard deviation, variance and co-efficient of ~~variates~~ variations

b) What will be standard deviation if 30 lbs is subtracted from each weight

c) 30 lbs is added in each of weight

d) if all values are divided by 10

e) if all values are multiplied by 10.

Soln:

Arranging: 120, 140, 150, 160, 180

$\Sigma x = 750$, $n = 5$

$\Sigma x^2 = 114500$

Now,

(a) standard deviation $(\sigma) = \sqrt{\left(\frac{\Sigma x^2}{n}\right) - \left(\frac{\Sigma x}{n}\right)^2}$

$$= \sqrt{\frac{114500}{5} - \left(\frac{750}{5}\right)^2}$$

$$= 20$$

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Again

Variance $= \sigma^2$

$= 20^2$

$= 100$

Variation, variability, consistency, accuracy, uniformity, homogenity

Calculate C.V.

Co-eff of variation (C.V) $= \frac{\sigma}{\bar{x}} * 100\%$

$= \frac{20}{\frac{\Sigma x}{n}} * 100\%$

$= \frac{20}{\frac{750}{5}} * 100\%$

$= 13.33\%$

Higher CV, less consistency
Lower CV, high consistency.

13.33% variation is occured due to the explained variable and remaining 86.67% variation is occured due to the unexplained variable/error

b) ⇒ $\Sigma x' = \Sigma x - 5*30$

$= 750 - 150$

$= 600$

(b, c) ⇓

Calculation table must be created during exam.

Note:-

If the constant value is either subtracted or added to the each observation, there will be no change in standard deviation.

d) → If all the values are divided by 10, standard deviation is also divided by 10.

$\therefore \sigma' = \frac{20}{10} = 2$

e) → If all the values are multiplied by 10, standard deviation is also multiplied by 10.

$\sigma' = 20 * 10 = 200$

Ans

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Q1 The arithmetic mean and standard deviation of 100 items are 60 and 10 respectively. In the process of calculation, it was found that an item 59 was incorrectly included as 99. Find correct mean and correct S.D. 08-03 PNS

Imp
Q.2 The combined mean and variance of salary of 250 worker of city A and B are 560 and 5497. The mean and variance of salary of 100 worker of city A are 650 and 121 respectively. find variance of salary in city B.

Q1 → Soln;

$n = 100$, $\bar{x} = 60$, $\sigma = 10$.

we have

$$\bar{x} = \frac{\Sigma x}{n}$$

$$\text{or } 60 = \frac{\Sigma x}{100}$$

$$\therefore \Sigma x = 6000$$

Now,

$$\sigma = \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2}$$

$$\text{or } \sigma^2 = \frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2$$

$$\text{or } 100 = \frac{\Sigma x^2}{100} - (60)^2$$

$$\text{or } 3700 = \frac{\Sigma x^2}{100}$$

$$\Sigma x^2 = 370000$$

$$\Sigma x' = 6000 - 99 + 59$$

$$= 5960$$

$$\bar{x}' = \frac{\Sigma x'}{n}$$

$$= \frac{5960}{100}$$

$$= 59.6$$

$$\Sigma x^{2\prime} = 370000 - (99)^2 + (59)^2$$

$$= 363680$$

$$\sigma' = 9.2$$

Ans

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(2) → Soln:

$n = 250.$

$\bar{x}_{AB} = 560$

$Var_{AB} = 5497$

$\sigma_{AB} = \sqrt{Var_{AB}}$

$\sigma_{AB} = 74.14$

$n_A = 100$ $\quad n_B = 150$

$\bar{x}_A = 650$

$Var_A = 121$

$\sigma_A = 11.$

Now, $\bar{x}_{AB} = \dfrac{n_A \bar{x}_A + n_B \bar{x}_B}{n_A + n_B}$

$\Rightarrow 560 = \dfrac{100 * 650 + 150 * \bar{x}_B}{250}$

$\Rightarrow \dfrac{140000 - 65000}{150} = \bar{x}_B$

$\therefore \bar{x}_B = 500$

$d_A = \bar{x}_A - \bar{x}_{AB}$, $d_B = \bar{x}_B - \bar{x}_{AB}$

$= 90$ $\quad = -60$

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08-03
PNS

we have

$$\sigma_{AB} = \sqrt{\frac{n_A(\sigma_A^2 + d_A^2) + n_B(\sigma_B^2 + d_B^2)}{n_A + n_B}}$$

$$\therefore \sqrt{5497} = \sqrt{\frac{100(121 + 8100) + 150(\sigma_B^2 + 3600)}{250}}$$

~~$137 4250 = 100\sigma_A^2 + 810000 + 18150 + 540000$~~

or,

$$\therefore 1374250 = 12100 + 810000 + 150\sigma_B^2 + 540000$$

$$\therefore 12150 = 150\sigma_B^2$$

$$\therefore \sigma_B^2 = 81$$

~~$\sigma_B = 9$~~

$$\therefore \text{Variance} = \sigma_B^2 = 81$$ Ans

IMP

Q.1 The lives of two model (A and B) of refrigerators in a recent survey are shown

Life (No. of yrs)	No. of refrigerators A	B
0-2	5	2
2-4	16	7
4-6	13	12
6-8	7	19
8-10	5	1
10-20	4	1

a) What is the average life cycle of each model of refrigerator
b) Which model has greater uniformity ?

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Imp

(2Q) An analysis of monthly wages paid to the workers 08-03 PNS
in two firm A and B belonging to same industry gives the following result.

	firm A	firm B
no. of workers	500	600
Average monthly wage	186	175
Variance of wages.	81	100

a) Which firm A or B has a larger wage bills?

b) In which firm A or B, is there greater variability in individual wages?

c) Calculate:
   i) average monthly wages.
   ii) the variance of distribution of wage of all worker in firm A and B taken together.

(Q.1)

→ Soln; Calculation table for refrigerator of model A

Life (No. of yrs)	No. of refrigerator A [f_A]	x	$f_A \cdot x$	x^2	$f_A x^2$
0-2	5	1	5	1	5
2-4	16	3	48	9	144
4-6	13	5	65	25	325
6-8	7	7	49	49	343
8-10	5	9	45	81	405
10-20	4	15	60	225	900
	$\Sigma f_A = N_A = 50$		$\Sigma f_A x = 272$		$\Sigma f_A x^2 = 2122$

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$$\text{Mean } (\bar{x}_A) = \frac{\Sigma f_A x}{\Sigma f_A}$$

$$= \frac{292}{50}$$

$$= 5.84$$

$$\sigma_A = \sqrt{\frac{\Sigma f_A x^2}{N_A} - \left(\frac{\Sigma f_A x}{N_A}\right)^2}$$

$$= \sqrt{\frac{2122}{50} - (5.84)^2}$$

$$= 3.58$$

$$C.V._A = \frac{\sigma_A}{\bar{x}_A} \times 100\% = \frac{3.58}{5.84} \times 100\% = 65.8\%$$

Again

Calculation table for refrigerator of model B.

life (No. of yrs)	No. of refrigerators f_B	x	$f_B \cdot x$	x^2	$f_B \cdot x^2$
0-2	2	1	2	1	2
2-4	7	3	21	9	63
4-6	12	5	60	25	300
6-8	19	7	133	49	931
8-10	1	9	9	81	81
10-20	1	15	15	225	225
	$\Sigma f_B = N_B = 42$		$\Sigma f_B \cdot x = 240$		$\Sigma f_B \cdot x^2 = 1602$

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$$\bar{x}_B = \frac{\Sigma f_B x}{\Sigma f_B} = \frac{240}{42} = 5.71$$

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$$\sigma_B = \sqrt{\frac{\Sigma f_B x^2}{N_B} - \left(\frac{\Sigma f_B x}{N_B}\right)^2}$$

$$= \sqrt{\frac{1602}{42} - (5.71)^2}$$

$$= 2.35$$

$$C.V._B = \frac{\sigma_B}{\bar{x}_B} * 100\% = \frac{2.35}{5.71} * 100\% = 41.15\%$$

a) → The average life cycle for each moder are 5.44 yrs for model A and 5.71 yrs for model B.

b) The model B has less c.v. i.e 41.15% due to which it has greater uniformity. Ans

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Q.2

→ soln;

Given:

	firm A	firm B
No. of workers.	500	600
Avg. monthly wages.	186	175
Variance of wages.	81	100

a) wage bill for firm A = 500 * 186 = 93000

wage bill for firm B = 600 * 175 = 105000

Thus, firm B has larger wage bills.

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(b) Given: $\bar{x}_A = 186$

$(V.A.R)_A = 81$,

$\therefore \sigma_A = \sqrt{81} = 9$

Also,

$C.V._A = \frac{\sigma_A}{\bar{x}_A} * 100\%$

$= \frac{9}{186} * 100\%$

$= 4.83\%$

Again

$(V.A.R)_B = 100$; $\bar{x}_B = 175$

$\therefore \sigma_B = \sqrt{100} = 10$

Now,

$C.V._B = \frac{\sigma_B}{\bar{x}_B} * 100\%$

$= \frac{10}{175} * 100\%$

$= 5.71\%$

Thus, In firm A, there is greater variability.

c) Given:

$n_A = 500$, $n_B = 600$

$\bar{x}_{AB} = \frac{n_A \cdot \bar{x}_A + n_B \cdot \bar{x}_B}{n_A + n_B}$

$= \frac{500 * 186 + 600 * 175}{500 + 600}$

$= 180$.

i) The combined average monthly wage is 180.

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Now,

$d_A = \bar{x}_A - \bar{x}_{AB} = 186 - 180 = 6$

$d_B = \bar{x}_B - \bar{x}_{AB} = 175 - 180 = -5$

So,

$$\sigma_{AB} = \sqrt{\frac{n_A(\sigma_A^2 + d_A^2) + n_B(\sigma_B^2 + d_B^2)}{n_A + n_B}}$$

$$= \sqrt{\frac{500 * (81 + 36) + 600 * (100 + 25)}{500 + 600}}$$

$$= 11.01$$

$(VAR)_{AB} = \sigma_{AB}^2 = (11.01)^2 = 121.22$.

∴ The variance of distribution of wage of all worker in firm A and B taken together is 121.22. Ans

081-08-04

Unit - 8:

Correlation and Regression

→ Association betn two variables.

Correlation:-

$$r = \frac{cov(x, y)}{\sqrt{var(x)}\sqrt{var(y)}}$$

$$r = \frac{n\Sigma xy - \Sigma x \cdot \Sigma y}{\sqrt{n\Sigma x^2 - (\Sigma x)^2} \cdot \sqrt{n\Sigma y^2 - (\Sigma y)^2}}$$

↳ range lies betn -1 to 1.

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- If $r = 0$, there is no correlation co-efficient.
- If $r = 1$, there is perfectly positive correlation.
- If $r = -1$, there is perfectly negative correlation.
- If $r = 0 - 0.5$; low degree
- If $r = 0.5 - 0.75$; moderately
- If $r = 0.75 - 0.99$; High degree.

* Co-efficient of determination
$= r^2$

Significant = evidence

* Standard Error:

$$S.E_r = \frac{1 - r^2}{\sqrt{n}}$$

* Probable Error:

$$P.E_r = 0.674 * \frac{1 - r^2}{\sqrt{n}}$$

→ If $|r| < P.E(r)$, there is no significant evidence of correlation.

→ If $|r| > 6 * P.E(r)$, there is evidence of correlation.

Q.

Expenses	7	10	9	4	11	5	3
Sales	12	14	13	5	15	7	9

Calculate correlation co-efficient of following data and also test the significant of correlation and co-efficient of determination.

↳ Soln.

Ctd →

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Expenses (x)	Sales (y)	xy	x^2	y^2
7	12	84	49	144
10	14	140	100	196
9	13	117	81	169
4	5	20	16	25
11	15	165	121	225
5	7	35	25	49
3	4	12	9	16
$\Sigma x = 49$	$\Sigma y = 70$	$\Sigma xy = 573$	$\Sigma x^2 = 401$	$\Sigma y^2 = 824$

Now,

$$\text{correlation co-eff } (r) = \frac{n\Sigma xy - \Sigma x \cdot \Sigma y}{\sqrt{n\Sigma x^2 - (\Sigma x)^2} \cdot \sqrt{n\Sigma y^2 - (\Sigma y)^2}}$$

$$= \frac{7 * 573 - 49 * 70}{\sqrt{7 * 401 - (49)^2} \cdot \sqrt{7 * 824 - (70)^2}}$$

$$= 0.97$$

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$$\text{Probable error, } PE(r) = 0.674 * \frac{1 - r^2}{\sqrt{n}}$$

$$= 0.674 * \frac{1 - (0.97)^2}{\sqrt{7}}$$

$$= 0.01$$

Here,

$|r| > 6 * P.E.(r)$, there is significance of correlation.

Again:

Co-eff of determination $= r^2 = (0.97)^2 = 0.94$

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*Regression:

↳ The simple regression eqn y on x is given by,

$$y = a + bx \quad \text{—①}$$

the value of constant 'a' and 'b' can be determined by fitting two normal equation by ordinary least square estimation.

$$\Sigma y = na + b\Sigma x \quad \text{—②}$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2 \quad \text{—③}$$

The estimated regression eqn of y on x is,

$$\hat{y} = a + bx$$

↳ The simple regression eqn x on y.

$$x = a + by \quad \text{—①}$$

$$\Sigma x = na + b\Sigma y \quad \text{—③}$$

$$\Sigma xy = a\Sigma y + b\Sigma y^2 \quad \text{—②}$$

$$\therefore \hat{x} = a + by$$

K=1, Simple regression
K=2, multiple regression

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* Residual = $y - \hat{y}$

TSS → Total Sum of Square
SSR → Sum of square due to Regression
SSE → Sum of Square due to Error

* Standard Error Estimate:

$$= \sqrt{\frac{SSE}{n-k-1}}$$

$$= \sqrt{\frac{\Sigma y^2 - a\Sigma y - b\Sigma xy}{n-k-1}}$$

$TSS = SSR + SSE$

$SSE = \Sigma(y - \hat{y})^2$

$SST = \Sigma(y - \bar{y})^2$

$SSR = \Sigma(\hat{y} - \bar{y})^2$

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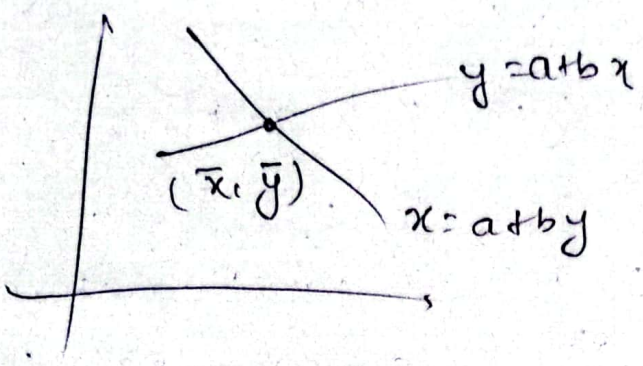

Q.1 Age 56 42 36 47 49 42 60 72 63 55
Pressure 147 125 118 128 145 140 155 160 149 150

a) find correlation co-efficient between age and pressure. Also find co-efficient of determination and test the significance.

b) fit two regression eqn.

c) Estimate the pressure when age is 45.

Soln;

Age (x)	Pressure (y)	xy	x^2	y^2
56	147	8232	3136	21609
42	125	5250	1764	15625
36	118	4248	1296	13924
47	128	6016	2209	16384
49	145	7105	2401	21025
42	140	5880	1764	19600
60	155	9300	3600	24025
72	160	11520	5184	25600
63	149	9387	3969	22201
55	150	8250	3025	22500
$\Sigma x = 1522$	$\Sigma y = 1417$	$\Sigma xy = 75188$	$\Sigma x^2 = 28348$	$\Sigma y^2 = 202493$

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(b) →

The simple regression eqn y on x is given by,

$$y = a + bx \quad -(1)$$

The value of constant 'a' and 'b' can be determined by fitting two normal eqn by ordinary least square estimation,

$$\Sigma y = na + b\Sigma x$$

$$\text{or, } 1417 = 10a + 522b \quad -(2)$$

$$\Sigma xy = a\Sigma x + b\Sigma x^2$$

$$\text{or, } 75188 = 522a + 28348b \quad -(3)$$

Solving (2) and (3), we get;

$$a = 83.75$$

$$b = 1.11$$

The estimated regression eqn y on x is

$$\hat{y} = 83.75 + 1.11x$$

Again,

The simple regression eqn x on y is given by,

$$x = a + by \quad -(4)$$

The value of constant 'a' and 'b' can be determined by fitting two normal eqn by ordinary least square estimation.

$$\Sigma x = na + b\Sigma y$$

$$\text{or } 522 = 10a + 1417b \quad -(5)$$

$$\Sigma xy = a\Sigma y + b\Sigma y^2$$

$$\text{or } 75188 = 1417a + 202493b \quad -(6)$$

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solving ⑤ and ⑥, we get,

$$a = -49.29$$

$$b = 0.71$$

∴ The estimated regression eqn x on y is,

$$\hat{x} = -49.29 + 0.71y$$

Alternative:

for regression eqn y on x;

$$b_{yx} = \frac{n\Sigma xy - \Sigma x \Sigma y}{n\Sigma x^2 - (\Sigma x)^2}$$

$$= \frac{10 * 75188 - 522 * 1419}{10 * 28398 - (522)^2}$$

$$= 1.11$$

$$a = \bar{y} - b\bar{x}$$

$$= \frac{\Sigma y}{n} - b * \frac{\Sigma x}{n}$$

$$= \frac{1417}{10} - 1.11 * \frac{522}{10}$$

$$= 83.75$$

∴ The estimated regression eqn y on x is,

$$\hat{y} = 83.75 + 1.11x$$

for regression eqn x on y:

$$b_{xy} = \frac{n\Sigma xy - \Sigma x \cdot \Sigma y}{n\Sigma y^2 - (\Sigma y)^2}$$

$$= \frac{10 * 75188 - 522 * 1417}{10 * 202493 - (1417)^2} = 0.71$$

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$$a = \bar{x} - b\bar{y}$$

$$= \frac{\Sigma x}{n} - b \cdot \frac{\Sigma y}{n}$$

$$= \frac{522}{10} - 0.71 * \frac{1417}{10}$$

$$= -48.4$$

∴ The estimated regression eqⁿ x on y is

$$\hat{x} = -48.4 + 0.71y$$

(c) →

Given $x = 45$

$\hat{y} = ?$

we have

$$\hat{y} = 83.75 + 1.11x$$

$$= 83.75 + 1.11 * 45$$

$$= 133.7$$

∴ The estimated pressure is 133.7 when age is 45.

Ans

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* Sampling Distribution and Estimation:

Sampling Distribution :

The probability distribution of a sample statistics is called sampling distribution.

Types of Sampling:

1) Probability Sampling.
2) Non-probability Sampling.

1. Probability Sampling: SRSWR/SRSWOR
   i) Simple Random Sampling (with and Without Replacement)
   ii) Systematic Random Sampling.
   iii) Stratified Random Sampling.
   iv) Cluster Random Sampling.
   v) Multistage Random Sampling.

2. Non-probability Sampling:
   i) Quota sampling
   ii) Purposive Sampling.
   iii) Judgement Sampling.
   iv) Convienience Sampling.
   v) Snowball Sampling.

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Q. A population consists of 5 member 2, 3, 6, 8, and 11. Consider all sample of size 2 that can be drawn without replacement from replacement from this population.

a) Find mean and variance of population.

b) Show that sample mean is unbiased estimator of population mean.

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c) find variance of sampling distribution of mean and also verify with formula. 08-09 PNS

$$V(\bar{x}) = \sigma^2_{\bar{x}} = \frac{\sigma^2}{n}\left(\frac{N-n}{N-1}\right)$$

d) Also show that standard deviation of sampling distribution of mean is less than population standard deviation.

↳ Soln;

Calculation table:

Y	Y^2
2	4
3	9
6	36
8	64
11	121

$\Sigma Y = 30$ $\Sigma Y^2 = 234$

$$\text{Mean}(\bar{Y}) = \frac{\Sigma Y}{n} = \frac{30}{5} = 6$$

$$\sigma = \sqrt{\frac{\Sigma Y^2}{n} - \left(\frac{\Sigma Y}{n}\right)^2}$$

$$= \sqrt{\frac{234}{5} - \left(\frac{30}{5}\right)^2}$$

$$= 3.289$$

$$V.A.R = \sigma^2$$

$$= 3.28^2$$

$$= 10.82$$

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The possible outcome of population of 5 members with a sample of size 2 is given by.

$$5C_2 = 10.$$

Let x denotes the sample population.

N^n

Calculation table

x	$\bar{x}$	$\bar{x}^2$
(2,3)	2.5	6.25
(2,6)	4	16
(2,8)	5	25
(2,11)	6.5	42.25
(3,6)	4.5	20.25
(3,8)	5.5	30.25
(3,11)	7	49
(6,8)	7	49
(6,11)	8.5	72.25
(8,11)	9.5	90.25
	$\Sigma\bar{x} = 60$	$\Sigma\bar{x}^2 = 400.5$

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Mean, $\bar{\bar{x}} = \frac{\Sigma\bar{x}}{n}$

$= \frac{60}{10} = 6$

$$\sigma_{\bar{x}} = \sqrt{\frac{\Sigma\bar{x}^2}{n} - \left(\frac{\Sigma\bar{x}}{n}\right)^2}$$

$$= \sqrt{\frac{400.5}{10} - 6^2}$$

$= 2.01$

$V.A.R(\bar{x}) = \sigma^2$

$= 2.01^2$

$= 4.04$

As, $\bar{\bar{x}} = \bar{x}$, the sample mean is unbiased estimator of population mean.

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c).

$$V.A.R(\bar{x}) = \sigma_{\bar{x}}^2 = 4.04$$

Now,

$$\frac{\sigma^2}{n}\left(\frac{N-n}{N-1}\right)$$

$$= \frac{10.32}{2}\left(\frac{5-2}{5-1}\right)$$

$$= 4.05 \approx 4.04$$

Thus,
the formula,

$$V(\bar{x}) = \sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}\left(\frac{N-n}{N-1}\right)$$

verified

d)

we have

$$\sigma_{\bar{x}} = 2.01$$

$$\sigma = 3.29$$

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Thus, the standard deviation of sampling distribution is less than population standard deviation

OR-10

Q. A population consists of 4 member 1, 2, 5 & 8. 08-09 PNS
Consider all sample of size 2, that can be drawn with replacement from this population.

a) find mean and variance of population.

b) Show that sample mean is unbiased estimator of pop^n mean

c) find variance of sampling distribution of mean and also verify with formula

$$V(\bar{x}) = \sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$$

d) Also show that standard deviation of sampling dist^n of mean is less than pop^n standard deviation.

Soln:

Calculation table:

Y	Y^2
1	1
2	4
5	25
8	64
$\Sigma Y = 16$	$\Sigma Y^2 = 94$

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$$\text{Mean}(\bar{Y}) = \frac{\Sigma Y}{n} = \frac{16}{4} = 4$$

$$\sigma = \sqrt{\frac{\Sigma Y^2}{n} - \left(\frac{\Sigma Y}{n}\right)^2}$$

$$= \sqrt{\frac{94}{4} - 16}$$

$$= 2.7386$$

$$V.A.R = \sigma^2 = 2.7386^2 = 7.4999$$

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The possible outcome of population of 4 member with a sample size of 2 is given by

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$$4^2 = 16$$

Let 'x' denotes the sample population

Calculation table:

x	$\bar{x}$	$\bar{x}^2$
(1,1)	1	1
(1,2)	1.5	2.25
(1,5)	3	9
(1,8)	4.5	20.25
(2,1)	1.5	2.25
(2,2)	2	4
(2,5)	3.5	12.25
(2,8)	5	25
(5,1)	3	9
(5,2)	3.5	12.25
(5,5)	5	25
(5,8)	6.5	42.25
(8,1)	4.5	20.25
(8,2)	5	25
(8,5)	6.5	42.25
(8,8)	8	64
	$\Sigma\bar{x} = 64$	$\Sigma\bar{x}^2 = 316$

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Mean, $\bar{\bar{x}} = \frac{\Sigma\bar{x}}{n}$

$= \frac{64}{16} = 4$

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$$\sigma_{\bar{x}} = \sqrt{\frac{\sum \bar{x}^2}{n} - \left(\frac{\sum \bar{x}}{n}\right)^2}$$

$$= \sqrt{\frac{316}{16} - 16}$$

$$= 1.9369$$

$$VAR(\bar{x}) = \sigma_{\bar{x}}^2$$

$$= (1.9369)^2$$

$$= 3.7499$$

As, $\bar{y} = \bar{x}$, the sample mean is unbiased estimator of population mean.

c)

$$\frac{\sigma^2}{n} = \frac{7.9999}{2}$$

$$= 3.749$$

$$= VAR(\bar{x})$$

So,

$$V(\bar{x}) = \sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$$ proved

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d)

$$\sigma_{\bar{x}} = 1.9369$$

$$\sigma = 2.7386$$

Thus, the standard deviation of sampling distribution is less than population standard deviation.

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081-08-10

08-10
Q.No 5

* Estimation:

↳ The statistical method of estimating the unknown population parameters from the sample observation drawn from the population is called estimation.

1. Point Estimation:

A point estimate of a population parameter in a single numerical value of statics that corresponds to that parameters.

2. Interval Estimation: (Fiducial limit)

↳ The procedure of constructing a random interval say (L, U) so, it has given probability $(1-\alpha)100\%$ of including the parameter θ is known as interval estimation. This consists in the determination of two constant L and U

i.e,

Imp $P(L<\theta<U) = 1-\alpha$

* Characteristics of good estimators:

1. Unbiasedness:

↳ Any estimator $\hat{\theta}$ of θ is said to be unbiased if it measure, what is expected to measure

i.e, $E(\hat{\theta}) = \theta$

2. Consistent estimator:

↳ An estimator $\hat{\theta}$ is said to be consistent estimator of population parameter of θ if as the sample size increase, the value of sample statistic becomes very close to value of population parameter.

i.e, $\lim_{n\to\infty} E(\hat{\theta}) \to \theta$

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3. Efficiency

↳ An estimator $\hat{\theta}$ is said to be efficient estimator of parameter θ if the variance of estimator $\hat{\theta}$ is least with comparison to variance of all other estimator of θ. Thus if $\hat{\theta}_1$ and $\hat{\theta}_2$ are consistent estimator of θ i.e.

$$var(\hat{\theta}_1) < var(\hat{\theta}_2) \text{ for all } n.$$

$$E = \frac{var(\hat{\theta}_1)}{var(\hat{\theta}_2)} < 1$$

4. Sufficient:

↳ A statistics is said to be sufficient estimator of parameter if it contain all the information in the sample about parameters.

Statistics	Standard Error
1. Mean (When σ known & pop^n infinite)	$S.E(\bar{x}) = \frac{\sigma}{\sqrt{n}}$
2. Mean (When σ known & pop^n finite)	$S.E(\bar{x}) = \frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}$
3. Mean (When σ unknown & pop^n infinite	$S.E(\bar{x}) = \frac{s}{\sqrt{n}}$
4. Mean (When σ unknown & pop^n finite	$S.E(\bar{x}) = \frac{s}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}$
5. Difference of Mean (when σ known)	$S.E(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$
6. Difference of mean (when σ unknown)	$S.E.(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

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7. Proportion (When popⁿ size is infinite)	$S.E.(p) = \sqrt{\frac{PQ}{n}}$
8. Proportion (When popⁿ size is finite)	$S.E.(p) = \sqrt{\frac{PQ}{n}} \sqrt{\frac{N-n}{N-1}}$
9. Difference of proportion	$S.E.(P_1-P_2) = \sqrt{\frac{P_1Q_1}{n_1} + \frac{P_2Q_2}{n_2}}$

* Critical Value of z.

Confidence $(1-\alpha)$	50%	68.27%	90%	95%	95.45%
Significance value	0.6745	1	1.645	1.96	2

98%	99%	99.73%
2.33	2.58	3

* Confidence Interval:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

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$$P\left[-z_{\alpha/2} \le \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} \le z_{\alpha/2}\right] = 1-\alpha$$

$$C.I. = [\bar{x} \pm z_{\alpha/2} \cdot S.E.(\bar{x})]$$

$$= \left[\bar{x} \pm z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}\right]$$

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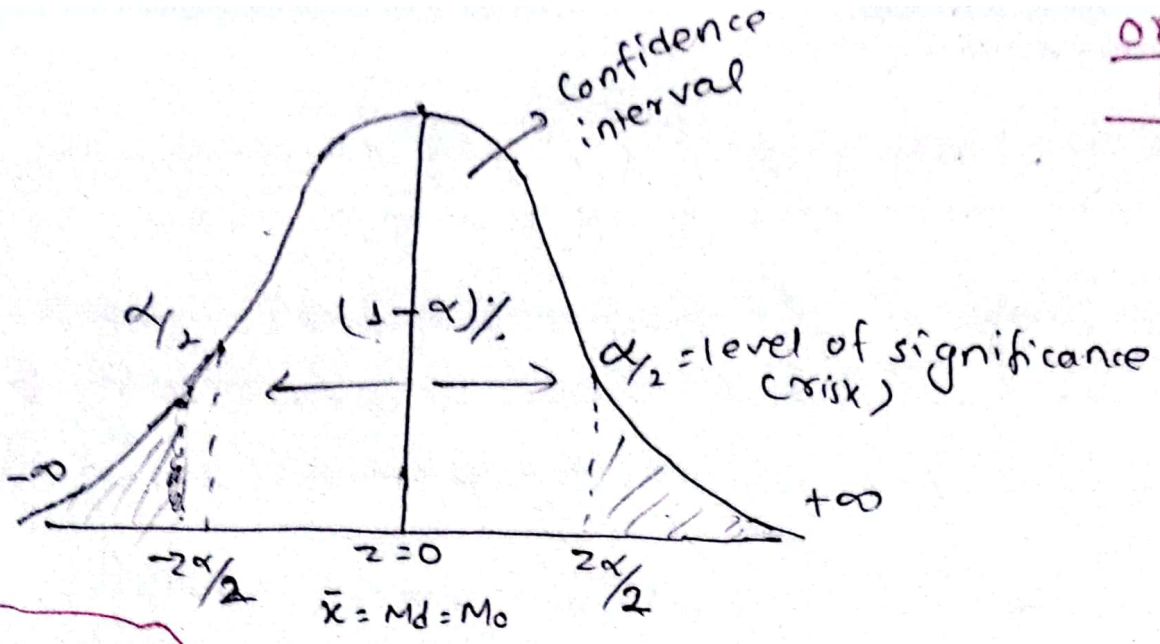

Z-test: $n \geq 30$

1. Confidence interval for single mean :-

$$C.I. = \left[\bar{x} \pm \frac{z\alpha}{2} \cdot \frac{\sigma}{\sqrt{n}} \right] \quad \text{(infinite)}$$

($\bar{x}$ ↓ sample mean.)

$$C.I. = \left[\bar{x} \pm \frac{z\alpha}{2} \cdot \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right] \quad \text{(finite)}$$

2. Confidence interval for mean difference :

$$C.I. = \left[(\bar{x}_1 - \bar{x}_2) \pm \frac{z\alpha}{2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \right] \quad \text{(infinite)}$$

$$C.I = \left[(\bar{x}_1 - \bar{x}_2) \pm \frac{z\alpha}{2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \right] \quad \text{(finite)}$$

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3. Confidence interval for single proportion

$$C.I. = pp \pm \frac{z\alpha}{2} \sqrt{\frac{pq}{n}} \quad \text{(infinite)}$$

$$C.I. = p \pm \frac{z\alpha}{2} \sqrt{\frac{pq}{n}} \quad \text{(finite)}$$

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08-10
PNS

4. Proportion difference:

$$C.I. = \left[(P_1 - P_2) \pm \frac{Z\gamma}{2} \cdot \sqrt{\frac{P_1Q_1}{n_1} + \frac{P_2Q_2}{n_2}}\right]$$

(Q.1) A sample of $n = 30$ meters showed a sample mean of $\bar{x} = 240$ and sample standard deviation is $s = 15$. find 90% and 15% confidence interval for mean.

(Q.2) A random sample of $n = 100$ articles selected from a batch of $N = 2000$ articles shows that the average diameter of article is $\bar{x} = 0.354$ with standard deviation $s = 0.048$. find 95% and 98% confidence interval for average of this batch of 2000 articles.

(Q.3)

	Brand A	Brand B
Mean	1120 $\bar{x}_1$	1062 $\bar{x}_2$
S.d	75 σ_1	82 σ_2
Number	100 n_1	100 n_2

Calcalate the 95% confidence interval.

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(Q.4) An examination of a sample of $n = 500$ pieces of pipes revealed that $x = 30$ were defective. find 98% confidence limits within which true proportion of defective item lies. $p = x/n$ $q = 1 - p$

(Q.5) In a random sample of $n = 300$ household in a city, $x = 223$ have computer. Set 95% limit to true value of proportion of city. $p = x/n$, $q = 1 - p$.

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Q.6 In a random sample 136 of 100 person gives flue vaccine experienced some discomfort. find 98% confidence interval for the proportion of person. 08-10 RNS

Q.7 It is claimed that both tea and coffee are equally popular in Ilam district. A random sample of 1200 person, 650 were regular consumer of tea. Estimate 95% confidence level. $p=0.5$, $Q=0.5$ $p = 650/1200$

Q.8 Random sample of 250 bolts manufactured by machine A and 200 bolts by B showed 24 and 10 is defective. Test that C.I for two population proportion at 95% level of confident.

Q.1) Ans

Soln;

Sample $(n) = 30$

sample mean $(\bar{x}) = 240$

Sample std deviation $(s) = 45$

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For 90% C.I.

$$C.I = \left[\bar{x} \pm z\frac{\alpha}{2} * \frac{s}{\sqrt{n}}\right]$$

$$= \left[240 \pm 1.645 * \frac{45}{\sqrt{30}}\right]$$

$$= [240 \pm 13.51]$$

$\therefore$ 90% confidence interval is. $[226.49, 253.51]$

Ctd →

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PNS

4 for 95% C.I:

$$CI = \left[\bar{x} \pm \frac{z\alpha}{2} * \frac{\sigma}{\sqrt{n}}\right]$$

$$= \left[240 \pm 1.96 * \frac{15}{\sqrt{30}}\right]$$

$$= [240 \pm 16.1]$$

∴ The 95% C.I is [223.9, 256.1] Ans:

Q.2) Ans

Soln;

sample (n) = 100

Total (N) = 2000

Mean ($\bar{x}$) = 0.354

Standar deviation (s) = 0.048

for 95% C.I

$$C.I = \left[\bar{x} \pm \frac{z\alpha}{2} * \frac{s}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}\right]$$

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$$= \left[0.354 \pm 1.96 * \frac{0.048}{\sqrt{100}} * \sqrt{\frac{2000-100}{2000-1}}\right]$$

$$= [0.354 \pm 0.009]$$

∴ The 95% C.I is [0.345, 0.363]

for 98% C.I.

$$C.I = \left[0.354 \pm 2.33 * \frac{0.048}{\sqrt{100}} * \sqrt{\frac{2000-100}{2000-1}}\right]$$

$$= [0.354 \pm 0.01]$$

∴ The 98% C.I is [0.344, 0.364] Ans

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Q.3) Ans

Soln;

Mean of Brand A ($\bar{x}_1$) = 1120

Mean of Brand B ($\bar{x}_2$) = 1062

S.D. of brand A (σ_1) = 75

S.D. of brand B (σ_2) = 82

sample of brand A (n_1) = 100

Sample of brand B (n_2) = 100

for 95% C.I:

$$C.I = \left[(\bar{x}_1 - \bar{x}_2) \pm \frac{z\alpha}{2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \right]$$

$$= \left[58 \pm 1.96 * \sqrt{\frac{75^2}{100} + \frac{82^2}{100}} \right]$$

$$= [58 \pm 21.78]$$

∴ The 95% CI is [36.22, 79.78] Ans

Q.4) Ans

Soln;

sample (n) = 500

defective pipes (x) = 30

Now,

$$\text{proportion } (p) = \frac{x}{n} = \frac{30}{500} = 0.06$$

Also,

$$q = 1 - p = 1 - 0.06 = 0.94$$

for 98% C.I:

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08-10
PNS

4. $C.I = \left[p \pm \frac{z\alpha}{2} \sqrt{\frac{pq}{n}} \right]$

$= \left[0.06 \pm 2.33 * \sqrt{\frac{0.06 * 0.94}{500}} \right]$

$= [0.06 \pm 0.02]$

∴ The 98% C.I is $[0.04, 0.08]$ Ans

Q.5) Ans

Soln;

Sample $(n) = 300$

no. of computers $(x) = 223$

household having

Now,

$p = \frac{x}{n} = \frac{223}{300} = 0.74$

Also,

$q = 1 - p = 1 - 0.74 = 0.26$

$p + q = 1$

For 95% C.I.

$C.I = \left[p \pm \frac{z\alpha}{2} \sqrt{\frac{pq}{n}} \right]$

$= \left[0.74 \pm 1.96 * \sqrt{\frac{0.74 * 0.26}{300}} \right]$

$= [0.74 \pm 0.04]$

∴ The 95% C.I is $[0.70, 0.78]$ Ans

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Q.6) Ans

08-10
PNS

Soln:

sample (n) = 400

no. of person feeling discomfort (x) = 136

Now.

$$p = \frac{x}{n} = \frac{136}{400} = 0.34$$

$$q = 1 - p = 1 - 0.34 = 0.66$$

For 98% C.I

$$C.I = \left[p \pm \frac{z\alpha}{2} \sqrt{\frac{pq}{n}} \right]$$

$$= \left[0.34 \pm 2.33 * \sqrt{\frac{0.34 * 0.66}{400}} \right]$$

$$= [0.34 \pm 0.05]$$

∴ The 98% C.I. is [0.29, 0.39] Ans

Q.7) Ans

Soln;

P = 0.5, Q = 0.5 } As both tea and coffee are equally popular

sample (n) = 1200

Regular customer of tea, (x) = 650

Now

$$p = \frac{x}{n} = \frac{650}{1200}$$

$$= 0.54$$

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4 For 95% CI

08-10
PNS

$$C.I = \left[p \pm z_{\frac{\alpha}{2}} \sqrt{\frac{pq}{n}} \right]$$

$$= \left[0.54 \pm 1.96 * \sqrt{\frac{0.5 * 0.5}{1200}} \right]$$

$$= [0.54 \pm 0.02]$$

S.E(p)

$\Rightarrow$

C.I =

∴ The 95% C.I. is $[0.52, 0.56]$ Ans

Q.8) Ans

Soln;

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081-08-11

08-11

PNJ

* T-test : $n<30$

1. Confidence interval for single mean:

$$t = \frac{\bar{x}-\mu}{s/\sqrt{n}} \sim t_{n-1}$$

$$C.I. = \left[\bar{x} \pm t_{\alpha/2,(n-1)} \cdot \frac{s}{\sqrt{n}}\right] \quad ; \quad s^2 = \frac{1}{n-1}\left[\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right]$$

$$C.I. = \left[\bar{x} \pm t_{\alpha/2,(n-1)} \cdot \frac{s}{\sqrt{n-1}}\right]$$

(Q.1) A random sample of size $n = 25$ showed a mean of $\bar{x} = 65$ inches with a standard deviation of $s = 2.5$ inches. Determine 98% confidence interval for mean.

fmp

(Q.2) A sample of pieces is taken and diameters are 1.01, 0.97, 1.03, 1.04, 0.99, 0.98, 0.99, 1.01 and 1.03. Find 99% confidence interval.

(Q.3) A random sample of 16 length values from normal population showed a mean of 41.5 inches and sum of square of deviation from this mean is equal to 135 inches. Show that whether if the mean of 43.5 inches for unknown population mean is reasonable from perspective of 95% confidence level.

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2. Confidence interval for mean difference.

$$\text{C.I.} = \left[(\bar{x}_1 - \bar{x}_2) \pm t_{\frac{\alpha}{2}, n_1+n_2-2} \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \right]$$

$$s_1^2 = \frac{1}{n_1 - 1}\left[\Sigma x_1^2 - \frac{(\Sigma x_1)^2}{n_1} \right]$$

$$s_2^2 = \frac{1}{n_2 - 1}\left[\Sigma x_2^2 - \frac{(\Sigma x_2)^2}{n_2} \right]$$

$$\text{C.I.} = \left[(\bar{x}_1 - \bar{x}_2) \pm t_{\frac{\alpha}{2}, n_1+n_2-2} \cdot s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \right]$$

$$s_p = \sqrt{\frac{1}{n_1+n_2-2}\left[\Sigma x_1^2 - \frac{(\Sigma x_1)^2}{n_1} + \Sigma x_2^2 - \frac{(\Sigma x_2)^2}{n_2} \right]}$$

Imp

(Q.1) Two different types of drugs D_1 and D_2 were administered on certain patients for increasing weight at interval of one week time period. Estimate 1% level of significant.

D_1	8	12	13	9	3	8	10	9	
D_2	10	8	12	15	6	11	12	12	

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(Q.2) Ans

Soln;

x	1.01	0.97	1.03	1.04	0.99	0.98	0.99	1.01	1.03
x^2	1.0201	0.9409	1.0609	1.0816	0.9801	0.9604	0.9801	1.0201	1.0609

$\Sigma x = 9.05$

$\Sigma x^2 = 9.1051$

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Now,

$$s = \sqrt{\frac{1}{n-1}\left[\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right]}$$

$$= \sqrt{\frac{1}{9-1}\left[9.1051 - \frac{(9.05)^2}{9}\right]}$$

$$= 0.0245$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{9.05}{9} = 1.0055$$

For 99% C.I.

$$C.I. = \left[\bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}\right]$$

$$= \left[1.0055 \pm 3.355 * \frac{0.0245}{\sqrt{9}}\right]$$

$$= [1.0055 \pm 0.0273]$$

∴ The 99% C.I is. [0.9782, 1.0328] Ans

(Q.3) Ans

$n = 16$

$\bar{x} = 41.5$

$\Sigma(x_i - \bar{x})^2 = 135$.

$s = \sqrt{\frac{1}{n-1}\Sigma(x_i - \bar{x})^2}$

Now,

$$s = \sqrt{\frac{1}{16-1} * 135}$$

$$= 3$$

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For 95% C.I.

$$C.I = \left[\bar{x} \pm t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n}}\right]$$

$$= \left[41.5 \pm 2.13 \times \frac{3}{\sqrt{16}}\right]$$

$$= [41.5 \pm 1.598]$$

The 95% CI is $[39.902, 43.098]$

Ans. the mean 43.5 does lie in 95% C.I, it is not reasonable from perspective of 95% confidence level.

(Q9) Ans

Soln; for drug D_1 | for drug D_2

$D_1(x_1)$	x_1^2	$D_2(x_2)$	x_2^2
8	64	10	100
12	144	8	64
13	169	12	144
9	81	15	225
3	9	6	36
8	64	11	121
10	100	12	144
9	81	12	144

$\Sigma x_1 = 72$ $\Sigma x_1^2 = 712$ $\Sigma x_2 = 86$ $\Sigma x_2^2 = 978$

$n_1 = 8$, $n_2 = 8$ $\bar{x}_1 = \frac{\Sigma x_1}{n_1} = \frac{72}{8} = 9$

$\bar{x}_2 = \frac{\Sigma x_2}{n} = \frac{86}{8} = 10.75$

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we have,

$$C.I. = \left[(\bar{x}_1 - \bar{x}_2) \pm t_{\frac{\alpha}{2}, n_1+n_2-2} \cdot S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}} \right]$$

Here,

$$S_p = \sqrt{\frac{1}{n_1+n_2-2} \left[\Sigma x_1^2 - \frac{(\Sigma x_1)^2}{n_1} + \Sigma x_2^2 - \frac{(\Sigma x_2)^2}{n_2} \right]}$$

$$= \sqrt{\frac{1}{8+8-2} \left[712 - \frac{72^2}{8} + 978 - \frac{86^2}{8} \right]}$$

$$= 2.89$$

So, for 1% significance

$$C.I = \left[(9 - 10.75) \pm t_{\alpha/2, 14} \cdot 2.89 \sqrt{\frac{1}{8} + \frac{1}{8}} \right]$$

$$= \left[-1.75 \pm 2.97 * 2.89 * \sqrt{\frac{1}{4}} \right]$$

$$= \left[-1.75 \pm 4.29 \right]$$

$$= -6.04,$$

∴ The 1% level of significant is $[-6.04, 2.54]$ Ans

(Q.1) Ans.

Soln;

$$n = 25, \quad \bar{x} = 65, \quad s = 25$$

For 98% C.I

$$C.I = \left[\bar{x} \pm t_{\alpha/2, n-1} \cdot \frac{s}{\sqrt{n-1}} \right]$$

$$= \left[65 \pm 2.492 * \frac{25}{\sqrt{24}} \right]$$

$$= [65 \pm 12.71]$$

∴ The 98% C.I is $[52.29, 77.71]$ Ans

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Q.6 Determine the minimum sample size required so that sample estimate lies within 10% of true value with 95% level of confidence when co-efficient of variation is 60%.

Solⁿ,

$$P(|\bar{x}-\mu|) \leq 0.10\mu = 0.95 \quad \text{---(1)}$$

$$\frac{P(|\bar{x}-\mu|)}{\sigma/\sqrt{n}} \leq z_{\alpha/2} = 0.95$$

or, $P(|\bar{x}-\mu|) \leq z_{\alpha/2} \cdot \sigma/\sqrt{n} = 0.95 \quad \text{---(2)}$

Equating (1) and (2),

$$0.1\mu = z_{\alpha/2} \cdot \sigma/\sqrt{n}$$

$$\therefore \sqrt{n} = \frac{z_{\alpha/2} \cdot \sigma}{0.1\mu}$$

$$\therefore n = \frac{z_{\alpha/2}}{0.1^2} \cdot \left(\frac{\sigma}{\mu}\right)^2$$

$$= \frac{1.96^2}{0.1^2} \times (0.60)^2$$

$$= 138.29 \approx 139$$

Ans

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08-10
PNS

* Hypothesis:

A tentative theory adopted to explain certain facts and guide in the investigation of others.

* Setting of hypothesis:

1. Null Hypothesis (H_0) :- $\bar{x} = \mu$
   [There is no significance difference bet^n sample mean & population mean].
2. Alternative Hypothesis (H_1) :- $\bar{x} \neq \mu$ [Two tail]
   [There is significance difference bet^n sample mean & population mean]
   $H_1 : \bar{x} > \mu$ [Right tail] } one tail.
   $H_1 : \bar{x} < \mu$ [left tail] }
3. Test statistics under H_0

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

4. Confidence interval :- $(1-\alpha)\% = 95\%$

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5. Level of significance :- $\alpha\% = 5\%$.

$$Z_{\alpha/2} = 1.96$$

6. Decision:-

If $Z_{cal} \geq Z_{tab}$ H_0 rejected

If $Z_{cal} < Z_{tab}$ H_0 accepted.

7. Conclusion.

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Types of error:

1) Type I error: It is the error of rejecting null hypothesis H_0 when it is true.

$\alpha = p(\text{Type I error})$

$= p(\text{Reject } H_0 \text{ when } H_0 \text{ is true})$

2) Type II error: It is the error of accepting null hypothesis H_0 when it is false

$\beta = p(\text{accept } H_0 \text{ when } H_1 \text{ is true})$.

power of test $(1-\beta) = p(\text{accepting } H_0 \text{ when } H_0 \text{ is true})$

P-value approach: The probability of obtaining a test statistics at least as extreme as one that was actually observed assuming H_0 is true is called p-value.

if $p < \alpha$; Reject H_0 otherwise accept H_0

if $p < \alpha$; Reject H_0;

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Z-test : $n \geq 30$

1. Test the significance of single mean :-

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

2. Test the significance of mean difference:

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$H_0 : \bar{x}_1 = \bar{x}_2$

$H_1 : \bar{x}_1 \neq \bar{x}_2$

$\bar{x}_1 > \bar{x}_2$

$\bar{x}_1 < \bar{x}_2$

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3. Test the significance of single proportion

08-16
PNS

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$$

H_0:

$$z = \frac{p - P}{\sqrt{\quad}}$$

4. Test the significance of population difference

$$z = \frac{p_1 - p_2}{\sqrt{\frac{p_1 q_1}{n_1} + \frac{p_2 q_2}{n_2}}}$$

(Q.) A sample of 400 students is found to have mean height of 170 cm. Can it be reasonably regarded as a sample from a large pop^n with mean height 169.5 cm and standard deviation 3.5 cm?

↳ soln:

$n = 400$

$\bar{x} = 170$

pop^n mean(μ) = 169.5

pop^n. s.d (σ) = 3.5

1) Null Hypothesis, $H_0: \bar{x} = \mu$

[It can be reasonably regarded as a sample from a large pop^n].

2) Alternative Hypothesis H_1 :- $\bar{x} \neq \mu$ (Two tail test).

[It cannot be reasonably regarded as a sample from a large pop^n].

3) Test statistics: Under H_0.

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{170 - 169.5}{3.5/\sqrt{400}} \quad \therefore z_{cal} = 2.85$$

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4) Confidence interval $(1-\alpha)\% = 95\%$

5) level of significance $(\alpha)\% = 5\%$

$$Z_{\alpha/2} = 1.96 \quad \therefore |Z_{tab}| = 1.96$$

6) Decision:

Here,

$$|Z_{cal}| \geq |Z_{tab}|;$$

So, H_0 is rejected.

7) Conclusion:

It cannot be reasonably regarded as a sample from a large pop^n.

Ans

(Q3) A manufacturer of ball pens claims that mean writing life of 400 pages with a standard deviation of 20 pages. A purchasing agent selects a sample of 100 pens and puts them for test. The mean writing life for the sample was 390 pages. Should the purchasing agent reject the manufacturer's claim at 5% level?

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↳ Soln;

$n = 100$

$\bar{x} = 390$

$\mu = 400$

$\sigma = 20$

$\alpha = 5\% = 0.05$

1) Null Hypothesis; H_0 :- $\mu = 400$

[The purchasing agent should not reject the manufacturer's claim]

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08-16
PNS

2) Alternative Hypothesis: $H_1: \mu \neq 400$

[The purchasing agent should reject the manufacturer's claim]

3) Test statistics: under H_0.

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{390 - 400}{20/\sqrt{100}}$$

$\therefore z_{cal} = -5$

$\therefore |z_{cal}| = 5.$

4) Confidence Interval $(1-\alpha)\,100\%$

$= (1 - 0.05)\,100\%$

$= 95\%$

5) level of significance $(\alpha)\% = 5\%$.

$z_{\alpha/2} = 1.96$

$\therefore |z_{tab}| = 1.96$

6) Decision:

Here,

$|z_{cal}| > |z_{tab}|$

So, H_0 is rejected.

7) Conclusion:

The purchasing agent should reject the manufacturer's claim. Ans

(Q3) A random sample of books worn by 40 combat soldiers in a desert region showed an average life of 1.08 yrs with standard deviation of 0.05 yrs. Under the standard conditions the boots are known to have an average life of 1.28 yrs. Is there reason to assert at 0.05 level of significance that use in desert causes

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the mean life of such boots to decrease?

↳ Soln;

$n = 40$, $\alpha = 0.05 = 5\%$

$\bar{x} = 1.08$

$s = 0.05$

$\mu = 1.28$

1) Null Hypothesis ; $H_0 : \mu = 1.28$
[There is reason to assert that use in desert causes the mean life of boots to decrease]

2) Alternative Hypothesis ; $H_1 : \mu < 1.28$ [left tail]
[There is no reason to assert that use in desert causes the mean life of boots to decrease].

3) Test statistics : Under H_0.

$$z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{1.08 - 1.28}{0.05/\sqrt{40}} = -25.29.$$

$\therefore |Z_{cal}| = 25.29$

4) Confidence Interval $(1-\alpha)100\%$

$= (1 - 0.05)100\%$

$= 95\%$.

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5) level of significance $= \alpha\% = 5\%$.

$z_\alpha = -1.645$

$\therefore |Z_{tab}| = 1.645$

6) Decision:

$|Z_{cal}| > |Z_{tab}|$

So, H_0 is rejected.

7) Conclusion:
There is no reason to assert that use in desert causes the mean life of boots to decrease.

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Q4 Does an average box of cereal contain more than 368 grams of cereal? A random sample of 35 boxes showed a sample mean of 372.5 gram. The company has specified σ to be 15 grams and distribution to be normal-[$\alpha = 0.05$]. 08-XX PNS

Q5 The Nepal bulb company claims that bulb has avg life of 15000 hrs with standard deviation of 1200 hrs. However, there has been complaint against this claim from regular customers that a sample of 60 bulbs has ~~claim~~ average life of only 13000 hrs Can you justify this claim at 2% level of significance?

Soln;

$\mu = 15000$

$\sigma = 1200$

$n = 60$

$\bar{x} = 13000$

$\alpha\% = 2\% = 0.02$

1) Null Hypothesis; $H_0 : \mu = 15000$
[The claim of company is correct].

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2) Alternative Hypothesis; $H_1 : \mu < 15000$ (left tail)
[The claim of company is incorrect].

3) Test statistics : Under H_0

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} = \frac{13000 - 15000}{1200/\sqrt{60}} = -12.909$$

$\therefore |Z_{cal}| = 12.909$

4) Confidence Interval $(1-\alpha)\,100\% = (1-0.02)\,100\%$
$= 98\%$

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5) level of significance $(\alpha)\% = 2\%$

$z_\alpha = -2.05$

$\therefore |z_{tab}| = 2.05$

6) Decision:

$|z_{cal}| > |z_{tab}|$

So, H_0 is rejected.

7) Conclusion:

The claim of company is incorrect.

Q.6

	X	Y
No. of bulbs	100	100
Mean	1300	1218
Standard deviation	82	93

Test whether there is any significant difference in mean life of two makes?

Soln

$n_1 = 100, \quad n_2 = 100$

$\bar{x}_1 = 1300, \quad \bar{x}_2 = 1218$

$s_1 = 82, \quad s_2 = 93$

1) Null Hypothesis; $H_0 : \bar{x}_1 = \bar{x}_2$

[There is no significant difference in mean life of two makes].

2) Alternative Hypothesis; $H_1 : \bar{x}_1 \neq \bar{x}_2$ [two tail test]

[There is significant difference in mean life of two makes]

3) Test Statistics: under H_0

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{1300 - 1218}{\sqrt{\frac{82^2}{100} + \frac{93^2}{100}}} = 4.19$$

$\therefore |z_{cal}| = 4.19.$

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4) Confidence Interval $(1-\alpha)100\% = 95\%$

5) level of significance $(\alpha)\% = 5\%$.

$$z_{\alpha/2} = 1.96$$

$$\therefore |z_{tab}| = 1.96$$

6) Decision:

$$|z_{cal}| > |z_{tab}|$$

So, H_0 is rejected.

7) Conclusion:

There is significant difference in mean life of two makes].

(Q.7)

	A	B
sample	40	50
Mean	37.2	32.1
S.D.	3.8	1.3

At $\alpha = 0.01$, should it conclude that brand A cars have higher average gas mileage than that of brand B.

Soln:

$n_1 = 40$, $n_2 = 50$ $\alpha = 0.01 = 1\%$.

$\bar{x}_1 = 37.2$, $\bar{x}_2 = 32.1$

$S_1 = 3.8$ $S_2 = 1.3$

1) Null Hypothesis; $H_0 : \bar{x}_1 = \bar{x}_2$

[It should not conclude that brand A cars have higher average gas mileage than that of brand B].

2) Alternative Hypothesis; $H_1 : \bar{x}_1 > \bar{x}_2$ (Right tail)

[It should conclude that brand A cars have higher average gas mileage than that of brand B].

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3) Test statistics : Under H_0

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{37.2 - 32.1}{\sqrt{\frac{3.8^2}{40} + \frac{4.3^2}{50}}} = 5.96$$

$\therefore |Z_{cal}| = 5.96$

4) Confidence Interval $(1-\alpha)100\%$

$= (1 - 0.01)100\%$

$= 99\%$

5) level of significance $(\alpha)\% = 1\%$

$Z_\alpha = 2.33$

$\therefore |Z_{tab}| = 2.33$

6) Decision :

$|Z_{cal}| > |Z_{tab}|$

so, H_0 is rejected.

7) Conclusion :

It should conclude that brand A cars have higher average gas mileage than that of brand B.

(Q.8) A coin is tossed 800 times and heads appears 480 times. Can you say that the coin is unbiased at 1% level of significance ?

Soln: $n = 800$, $x = 480$

$p = \frac{x}{n} = \frac{480}{800} = 0.6$

$P = 0.5$, $Q = 0.5$

$q = 1 - p = 1 - 0.6 = 0.4$

Correct: $z = \frac{p - P}{\sqrt{\frac{PQ}{n}}}$

$\alpha\% = 1\% = 0.01$

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2) Null Hypothesis; $H_0: P = P = 0.5$ i.e. $P = 0.5$ [two tail] 08-16 PNS
[The coin is unbiased]

2) Alternative Hypothesis; $H_1: P \neq 0.5$
[The coin is baaiased]

3) Test statistics: Under H_0.

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.6 - 0.5}{\sqrt{\frac{0.5 \times 0.5}{800}}} = 5.65$$

$\therefore |z_{cal}| = 5.65$

4) Confidence interval $(1-\alpha)100\% = 99\%$.

5) level of significance $(\alpha)\% = 1\%$.

$$z_{\alpha/2} = 2.58$$

$\therefore |z_{tab}| = 2.58$

6) Decision:

$|z_{cal}| > |z_{tab}|$

So, H_0 is rejected.

7) Conclusion:
We can say that the coin is biased.

Q.9) The head teacher in school claimed that at least 98% of the students submit their assignment on time taking sample of 250 students, 15 weren't found submitting assignment in whole academic year. Test his claim at 10% level of significance.

Soln;

$n = 250$, $p = 0.98$, $\alpha = 10\% = 0.1$

$x = 235$ $Q = 1 - P = 0.02$

$\therefore p = \frac{x}{n} = \frac{235}{250} = 0.94$

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08-16
PNS

1) Null Hypothesis; H_0 : $P = 0.98$.
[The claim of headteacher is correct].

2) Alternative Hypothesis; H_1 : $P < 0.98$ (left tail).
[The claim of headteacher is incorrect].

3) Test statistics: Under H_0

$$z = \frac{p - P}{\sqrt{\frac{PQ}{n}}} = \frac{0.94 - 0.98}{\sqrt{\frac{0.98 * 0.02}{250}}} = -4.517$$

$\therefore |Z_{cal}| = 4.517$

4) Confidence interval $(1-\alpha)100\% = 90\%$

5) Level of significance $(\alpha)\% = 10\%$

$z_\alpha = -1.28$

$\therefore |Z_{tab}| = 1.28$

6) Decision:

$|Z_{cal}| > |Z_{tab}|$

So, H_0 is rejected.

7) Conclusion:
The claim of headteacher is incorrect.

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Q.10) A sample of 600 persons selected randomly from a large city gives the result that males are 53%. Is there reason to doubt the hypothesis that males and females are in equal number in city?

Ans

Soln;

$n = 600$.

$p = 0.53$ $\quad q = 1 - p = 0.47$

$P = 1/2 = Q$.

$H_0 : p = P$
$H_1 : p \neq P$

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Q.16 A coin is tossed 400 times and it turns up head 216 times. Discuss ~~beta~~ whether the coin is unbiased. Hints:

08-066
PNS

$x = 216, \quad n = 400 \quad p = \frac{216}{400} \quad q = 1 - p \quad P = Q = 0.5$

$$z = \frac{p - P}{\sqrt{\frac{pq}{n}}}$$

Q.17 In a random sample of 400 persons from a large pop^n, 120 are females. Can it be said that males and females are in ratio 5:3 in the pop^n? Use 10% level of significance.

$n = 400$

$p = \frac{120}{400}$, $q = 1 - p$

$P = 5/8$

$Q = 2$

$H_0 : P = 5/8$

$H_1 : P \neq 5/8$

Odds factors

$A : B = \frac{A}{A+B}$

$B : A = \frac{B}{A+B}$

Q.18 A developer has claimed that at least 98% of software which he supplied to a tribhuvan university conformed to specification an examination of a sample of 500 software revealed that 30 were defective. Test the claim at a significance level of 0.01.

$p = 0.98 \quad Q = 1 - P = 0.02$

$n = 500 \quad p = \frac{470}{500}$

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08-16
PNS

Q.11 A member of public interest group concerned with environment pollution asserts at public hearing that fewer than 60% of industrial plants in this area are complying with air pollution standards. The officials sample 60 plants and find that 33 are complying with air pollution standard. Is the asserting by member of public interest group a valid one? [Test hypothesis at 0.02 significance level.] P=0.6

Q.12 A random sample of 100 pen drive selected from a batch of 2000 pen drive shows that average thickness of pen drive is 0.354 with s.d. 0.048. Are the sample from a lot having average thickness 0.35?

↳ soln;

$\bar{x} = 0.354$

$s = 0.048$

$n = 100$

$N = 2000$

$\mu = 0.35$

1) Null Hypothesis; $H_0: \bar{x} = \mu$

[The sample are from a lot having average thickness 0.35].

2) Alternative Hypothesis; $H_1: \bar{x} \neq \mu$. [two tail]

[The sample arenot from a lot having avg. thickness 0.35].

3) Test statistics: under H_0

$$z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}} = \frac{0.354 - 0.35}{\frac{0.048}{\sqrt{100}}\sqrt{\frac{2000-100}{2000-1}}} = 0.85$$

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$\therefore |Z_{cal}| = 0.885$

4) Confidence interval $(1-\alpha)\% = 95\%$.

5) level of significance $(\alpha)\% = 5\%$.

$$Z_{\alpha/2} = 1.96$$

$$\therefore |Z_{tab}| = 1.96$$

6) Decision:

$$|Z_{cal}| < |Z_{tab}|$$

So, H_0 is accepted.

7) Conclusion:

The sample are from a lot having average thickness 0.35. Ans.

(Q.13) If the mean breaking strength of a copper wire is 575 lbs with s.d. of 8.3 lbs. How many large samples must be used in order that there be one chance in 100 that the mean breaking strength of sample is less than 572 lbs? $\bar{x} = np$ $p = 1/100$

(Q.14) The quality control manager at light bulb factory needs to determine whether the mean life of large shipment of light bulbs is equal to 375 hours. The pop^n s.d. is 100 hrs. A random sample of 100 light bulbs indicates a sample mean 350 hrs. At the 0.05 level of significance is there evidence that the mean life is different from 375 hours? Use P value method to draw conclusion.

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t-test: $n<30$

1. Test the significance of single mean:

$$t = \frac{\bar{x}-\mu}{s/\sqrt{n}} \sim t_{n-1}$$

biased

$$t = \frac{\bar{x}-\mu}{s/\sqrt{n-1}}$$

$$s^2 = \frac{1}{n-1}\left[\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right]$$

2. Test the significance of mean difference.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} \sim t_{n_1+n_2-2}$$

$$s_p^2 = \frac{1}{n_1+n_2-2}\left\{\Sigma x_1^2 - \frac{(\Sigma x_1)^2}{n_1} + \Sigma x_2^2 - \frac{(\Sigma x_2)^2}{n_2}\right\}$$

biased

$$s_p^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1+n_2-2}$$

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(Q.15) In a random sample of 500 the mean is found to be 20. In another independent sample of 400, the mean is 15. Could the sample have been drawn from the sample popⁿ with s.d. 4?

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Related to t-test

08-067 PNS

(Q.1) A company claims that mean life time of its electric light bulbs is 28 months. A random sample of 10 bulbs have following life month 24, 26, 32, 28, 20, 20, 23, 31, 30, & 43. Test the claim of company at 5% level of significance.

Soln;

$\mu = 28$

$n = 10$

$\alpha = 5\% = 0.05$

x	x^2
24	576
26	676
32	1024
28	784
20	400
20	400
23	529
31	1156
30	900
43	1849
$\Sigma x = 280$	$\Sigma x^2 = 8294$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{280}{10} = 28$$

$$s = \sqrt{\frac{1}{n-1}\left[\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right]}$$

$$= \sqrt{\frac{1}{10-1}\left[8294 - \frac{280^2}{10}\right]}$$

$$= 7.102$$

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1) Null Hypothesis; $H_0 : \mu = 28$

[The claim of company is correct.]

2) Alternative Hypothesis; $H_1 : \mu \neq 28$ [two tail]

[The claim of company is incorrect.]

3) Test statistics; under H_0

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{28-28}{7.102/\sqrt{10}} = 0$$

$\therefore |t_{cal}| = 0.$

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08-17
PNS

4) Confidence Interval $(1-\alpha)\% = 95\%$

5) level of significance $(\alpha)\% = 5\%$

6) Degree of freedom $= n-1 = 9$

1.

$$t_{\alpha/2, n-1} = 2.262$$

$$\therefore |t_{tab}| = 2.262$$

7) Decision:

$$|t_{cal}| < |t_{tab}|$$

so, H_0 is accepted

8) Conclusion:

The claim of company is correct that mean life time of its electric bulbs is 28 months. Ans

2

(9.2)

D_1	8	12	13	9	3	8	10	9
D_2	10	8	12	15	6	11	12	12

From the above observations, can you conclude that the second drug is more effective in increasing weight at 1% level of significance?

Soln;

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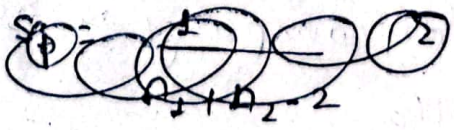
$D_1(x_1)$	x_1^2
8	64
12	144
13	169
9	81
3	9
8	64
10	100
9	81

$$\Sigma x_1 = 72, \quad \Sigma x_1^2 = 712$$

$$\bar{x}_1 = \frac{\Sigma x_1}{n_1} = \frac{72}{8} = 9$$



$\alpha = 1\% = 0.01$

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08-17
PNJ

$D_2(x_2)$	x_2^2
10	100
8	64
12	144
15	225
6	36
11	121
12	144
12	144

$\Sigma x_2 = 86, \quad \Sigma x_2^2 = 978$

$\bar{x}_2 = \frac{\Sigma x_2}{n_2} = \frac{86}{8} = 10.75$

$$S_p = \sqrt{\frac{1}{n_1+n_2-2}\left[\Sigma x_1^2 - \frac{(\Sigma x_1)^2}{n_1} + \Sigma x_2^2 - \frac{(\Sigma x_2)^2}{n_2}\right]}$$

$$= \sqrt{\frac{1}{8+8-2}\left[712 - \frac{(72)^2}{8} + 978 - \frac{86^2}{8}\right]}$$

$= 2.89$

1) Null Hypothesis ; $H_0: \bar{x}_1 = \bar{x}_2$
[Both drugs are equally effective]

2) Alternative Hypothesis ; $H_1: \bar{x}_1 < \bar{x}_2$ [left tail]
[Second drug is more effective in increasing weight].

3) Test statistics : Under H_0

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{S_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{9 - 10.75}{\sqrt{2.89^2\left(\frac{1}{8} + \frac{1}{8}\right)}} = -1.21$$

$\therefore |t_{cal}| = 1.21$

4) Confidence interval $(1-\alpha)\% = 99\%$

5) level of significance $(\alpha)\% = 1\%$.

6) Degree of freedom $= n_1 + n_2 - 2 = 14$

$t_{\alpha, n_1+n_2-2} = 2.624$

$\therefore |t_{tab}| = 2.624$

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7) Decision:

08-17
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$$|t_{cal}| < |t_{tab}|$$

$\therefore$ H_0 is accepted.

8) Conclusion:

We cannot conclude that the second drug is effective in increasing weight more.

Q. A manufacturer intends that his electric light bulbs have a life of 1000 hours. He test a sample of 20 bulbs, draw at random from a batch and discover that mean life of sample bulbs is 990 hrs with s.d. 22 hrs. Does this signify that batch is not up to the standard.

Soln;

$\mu = 1000$

$n = 20$

$\bar{x} = 990$

$s = 22$.

1) Null Hypothesis; $H_0 : \mu = 1000$

[It signifies that batch is ~~not~~ up to the std].

2) Alternative Hypothesis; $H_1 : \mu < 1000$ [left tail]

[It ~~does not~~ signifies that batch is not up to the std]

3) Test statistics: under H_0.

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n-1}}} = \frac{990 - 1000}{\frac{22}{\sqrt{20-1}}} \simeq -1.98$$

$\therefore |t_{cal}| = 1.98$

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4) Confidence Interval (1-α)% = 95%

5) level of significance (α)% = 5%

6) Degree of freedom = n-1 = 19

$t_{\alpha, n-1} = 1.729$

$\therefore |t_{tab}| = 1.729$

7) ~~Conclusion~~ Decision

$|t_{cal}| > |t_{tab}|$

$\therefore H_0$ is rejected

8) Conclusion:

This signifies that batch is not up to the standard

Ans

03-03-16

Imp

* Paired test:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} \sim t_{n-1}$$

Before, after → comparison → use paired t-test

$$\bar{d} = \frac{\Sigma d}{n}, \quad d = X - Y \text{ or } Y - X$$

$$s_d^2 = \frac{1}{n-1}\left[\Sigma d^2 - \frac{(\Sigma d)^2}{n}\right]$$

Imp

Q1 Sales executed by them before and after training are listed below:

before	23	20	19	21	18	20	22	49
after	25	25	24	24	22	23	25	21

Do these data indicate that training has contributed to their performance?

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Q.2 A drug is given to 10 patients, one the increments in their blood pressure were recorded to be 8, 10, -2, 0, 5, -1, 9, 12, 6, 5. Test at 5% level of significance whether the drug is effective for either increasing or decreasing the blood pressure?

Ans

soln;

$\alpha = 5\% = 0.05$

Calculation table

d	d^2
8	64
10	100
-2	4
0	0
5	25
-1	1
9	81
12	144
6	36
5	25
$\Sigma d = 52$	$\Sigma d^2 = 480$

$$\bar{d} = \frac{\Sigma d}{n} = \frac{52}{10} = 5.2$$

$$S_d^2 = \frac{1}{n-1}\left[\Sigma d^2 - \frac{(\Sigma d)^2}{n}\right]$$

$$= \frac{1}{10-1}\left\{480 - \frac{52^2}{10}\right\}$$

$$= 23.28$$

$$S_d = \sqrt{23.28} = 4.82$$

1) Null Hypothesis; $H_0: \mu_x = \mu_y$

[The drug is not effective for either increasing or decreasing the blood pressure]

2) Alternative Hypothesis; $H_1: \mu_x \neq \mu_y$ (two tail)

[The drug is effective for either increasing or decreasing the blood pressure]

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2) Test statistics : under H_0

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} = \frac{5.2}{\frac{4.82}{\sqrt{10}}} = 3.41$$

$\therefore |t_{cal}| = 3.41$

4) Confidence Interval $(1-\alpha)\% = 95\%$

5) level of significance $(\alpha)\% = 5\%$

6) Degree of freedom : $n-1 = 9$

$t_{\alpha/2, n-1} = 2.262$

$\therefore |t_{tab}| = 2.262$

7) Decision:

$|t_{cal}| > t_{tab}$

So, H_0 is rejected.

8) Conclusion:

The drug is effective for either increasing or decreasing the blood pressure.

(Q.1) → soln

Calculation table

Before (x)	After (y)	$d = y - x$	d^2
23	25	2	4
20	25	5	25
19	24	5	25
21	24	3	9
18	22	4	16
20	23	3	9
18	25	7	49
17	21	4	16
		$\Sigma d = 33$	$\Sigma d^2 = 153$

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$$\bar{d} = \frac{\Sigma d}{n} = \frac{33}{8} = 4.125$$

$$s_d^2 = \frac{1}{n-1}\left[\Sigma d^2 - \frac{(\Sigma d)^2}{n}\right]$$

$$= \frac{1}{8-1}\left[153 - \frac{(33)^2}{8}\right]$$

$$= 2.41$$

$\therefore s_d = 1.55$

1) Null Hypothesis, $H_0 : \mu_x = \mu_y$
[The training has not contributed to their performance]

2) Alternative Hypothesis ; $H_1 : \mu_x < \mu_y$ [left tail]
[The training has contributed to their performance]

3) Test statistics : under H_0

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} = \frac{4.125}{1.55/\sqrt{8}} = 7.52$$

$\therefore |t_{cal}| = 7.52$

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4) Confidence Interval $(1-\alpha)\% = 95\%$

5) level of significance $(\alpha)\% = 5\%$.

6) Degree of freedom : $n-1 = 7$

$t_{\alpha, n-1} = 1.895$

$\therefore |t_{tab}| = 1.895$

7) Decision :

$|t_{cal}| > |t_{tab}|$

so, H_0 is rejected.

8) Conclusion :
The training has contributed to their performance

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χ^2-test (chi-square test)

08-18
PNS

1) Null Hypothesis (H_0) :- There is no significance relationship between them (independence).

2) Alternative (H_1) :- There is significance association betn them (dependence).

3) Test statistics :

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$; O = observed, E = Expected.

$$E = \frac{R.T * C.T.}{N}$$; R.T = Row Total, C.T = Column Total.

or,

$$\chi^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)}$$

4) Critical region $(1-\alpha)\% = 95\%$

5) Level of significance $(\alpha)\% = 5\%$.

6) Degree of freedom : $(r-1).(c-1)$

7) Decision :

If $\chi^2_{cal} > \chi^2_{tab}$; H_0 rejected

If $\chi^2_{cal} < \chi^2_{tab}$; H_0 accepted

Q. Two group of 100 people each were taken for testing the use of vaccine 15 persons contracted the disease out of inoculated person, while 25 contracted disease in other group. Test efficiency of vaccine using χ^2-value at 5% level of significance.

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03-18
QNS

Soln;

	Present	Absent	Total
Inoculated	a=15	b=85	a+b=100
not inoculated	c=25	d=75	c+d=100
	a+c=40	b+d=160	N=200

1) Null Hypothesis (H0):

There is no significance relationship between inoculated and not inoculated.

2) Alternative Hypothesis (H1):

There is significance association between inoculated and not inoculated.

3) Test statistics:

$$\chi^2 = \frac{200\,(15*75 - 85*25)^2}{(100)*100*40*160}$$

$$= 3.125$$

$$|\chi^2_{cal}| = 3.125$$

4) Critical region : $(1-\alpha)\% = 95\%$

5) level of significance : $(\alpha)\% = 5\%$ → $|\chi^2_{tab}| = 3.841$

6) Degree of freedom : $(r-1)(c-1) = (2-1)(2-1) = 1$

7) Decision:

5 →

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08-08-23

08-23
PNS

*Probability

Probability : $\frac{\text{no. of favorable cases}}{\text{total no. of cases}}$

Permutation : $^{n}P_{x} = \frac{n!}{(n-x)!}$

Combination : $^{n}C_{x} = \frac{n!}{(n-x)!x!}$

(Q.1) In how many different ways can 3 (x) of 20 (n) laboratory assistants be choosen to assist with an experiments? Combination

(Q.2) In how many different ways can one make a first, second, third, and fourth choice among 12 firms leasing construction equipment? permutation

(Q.3) In a company, out of 10 condidates 7 men and 3 women apply for two vacancies, what is the probability that

a) both men are selected $^{7}C_{2}/^{10}C_{2}$

b) both women are selected $^{3}C_{2}/^{10}C_{2}$

c) one men and one women are selected $\frac{^{7}C_{1} * {}^{3}C_{1}}{^{10}C_{2}}$

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(Q.4) A bag contains 8 red, 4 white and 5 black balls. Three balls are drawn randomly from a bag. find the probability that

a) all red $\frac{^{8}C_{3}}{^{17}C_{3}}$

b) 2 red and 1 white $\frac{^{8}C_{2} * {}^{4}C_{1}}{^{17}C_{3}}$

c) 2 red and 1 other $\frac{^{8}C_{2} * {}^{9}C_{1}}{^{17}C_{3}}$

d) all colour balls. $\frac{^{8}C_{1} * {}^{4}C_{1} * {}^{5}C_{1}}{^{17}C_{3}}$

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Q.5 There are 3 scientist, 2 doctor and 4 professors. A committee of 3 from among them is to be formed. find the probability that. 078-23 PNS

a) all professor.

b) 1 scientist and 2 doctor

c) one of each kind.

d) 1 professor and 2 other.

e) no scientist.

f) at least one doctor.

a) $\frac{^4C_3}{^9C_3}$ b) $\frac{^3C_1 * ^2C_2}{^9C_3}$

c) $\frac{^3C_1 * ^2C_1 * ^4C_1}{^9C_3}$

d) $\frac{^4C_1 * ^5C_2}{^9C_3}$

e) $\frac{^6C_3}{^9C_3}$

f) $\frac{^2C_1 * ^9C_2}{^9C_3}$

Q.6 In an organization, there are 2 civil engineers, 4 IT officers, and 2 accountants. A committee of four members has formed. what is the probability that a committee contains.

a) 1 civil and 3 IT officers.

b) no civil.

c) at least one IT officers

a) $\frac{^2C_1 * ^8C_3}{^8C_4}$

b) $\frac{^6C_4}{^8C_4}$

c) $\frac{^4C_1 * ^8C_3}{^8C_4}$

Q.7 From a group of 3 Nepalese, 4 Indian and 5 American a subcommittee of four persons are selected randomly. find the probability that subcommittee will consists of.

a) 3 Nepalese and 1 Indian.

b) 1 Nepalese and 1 Indian and 2 American

c) 1 American.

a) $\frac{^3C_3 * ^4C_1}{^{12}C_4}$

b) $\frac{^3C_1 * ^4C_1 * ^5C_2}{^{12}C_4}$

c) $\frac{^5C_4}{^{12}C_4}$

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08-23

08-23

PNS

(Q3) soln

Total no. of candidate = 10

Total no. of men = 7

Total no. of women = 3

Total no. of vacancies = 2

a) P(both men) = $\frac{^7C_2}{^{10}C_2}$

$= \frac{7!}{(7-2)!\,2!} * \frac{(10-2)!*2!}{10!}$

$= \frac{7}{15}$

b) P(both women) = $\frac{^3C_2}{^{10}C_2}$

$= \frac{3!}{(3-2)!*2!} * \frac{(10-2)!*2!}{10!}$

$= \frac{1}{15}$

c) P(one men and one women) = $\frac{^7C_1 * {}^3C_1}{^{10}C_2}$

$= \frac{\frac{7!}{(7-1)!*1!} * \frac{3!}{(3-1)!*1!}}{\frac{10!}{(10-2)!*2!}}$

$= \frac{7}{15}$

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Q. (Q.4) soln;

Total no. of balls = 17

No. of red balls = 8

No. of white balls = 4

No. of black balls = 5

No. of balls drawn = 3

a) P(all red) = $\frac{{}^8C_3}{{}^{17}C_3} = \frac{8!}{(8-3)! * 3!} * \frac{(17-3)! * 8!}{17!}$

$= 7/85$

b) P(2 red and 1 white) = $\frac{{}^8C_2 * {}^4C_1}{{}^{17}C_3}$

$= \frac{\frac{8!}{(8-2)! * 2!} * \frac{4!}{(4-1)! * 1!}}{\frac{17!}{(17-3)! * 3!}}$

$= \frac{14}{85}$

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c) P(2 red and 1 other) = $\frac{{}^8C_2 * {}^9C_1}{{}^{17}C_3}$

$= \frac{\frac{8!}{(8-2)! * 2!} * \frac{9!}{(9-1)! * 1!}}{\frac{17!}{(17-3)! * 3!}}$

$= \frac{63}{170}$

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08-23
PNS

d) P(all colour ball) = $\frac{{}^8C_1 * {}^4C_1 * {}^5C_1}{{}^{17}C_3}$

$$= \frac{\frac{8!}{(8-1)!*1!} * \frac{4!}{(4-1)!*1!} * \frac{5!}{(5-1)!*1!}}{\frac{17!}{(17-3)!*3!}}$$

$$= \frac{4}{17} \quad \text{Ans}$$

(Q.5) → soln;

Total no. of committee member = 9

No. of scientist = 3

No. of doctor = 2

No of professor = 4

No. of committee members to be formed = 3

a) P(all professor) = $\frac{{}^4C_3}{{}^9C_3} = \frac{4!}{(4-3)!*3!} * \frac{(9-3)!*3!}{9!}$

$$= 1/21$$

b) P(1 scientists and 2 doctors) = $\frac{{}^3C_1 * {}^2C_2}{{}^9C_3}$

$$= \frac{\frac{3!}{(3-1)!*1!} * \frac{2!}{(2-2)!*2!}}{\frac{9!}{(9-3)!*3!}}$$

$$= 1/28$$

c) P(1 of each kind) = $\frac{{}^3C_1 * {}^2C_1 * {}^4C_1}{{}^9C_3}$

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08-23
PNS

$$= \frac{\frac{3!}{(3-1)!*1!} * \frac{5!}{(2-1)!*1!} * \frac{4!}{(4-1)!*1!}}{\frac{9!}{(9-3)!*3!}}$$

$$= 2/7$$

d) P(1 professor and 2 other) $= \frac{{}^4C_1 * {}^5C_2}{{}^9C_3}$

$$= \frac{\frac{4!}{(4-1)!*1!} * \frac{5!}{(5-2)!*2!}}{\frac{9!}{(9-3)!*3!}}$$

$$= 10/21$$

e) P(No scientist) $= \frac{{}^6C_3}{{}^9C_3} = \frac{6!}{(6-3)*3!} * \frac{(9-3)!*3!}{9!}$

$$= 5/21$$

f) P(at least 1 doctor) $= \frac{{}^2C_1 * {}^9C_2}{{}^9C_3}$

$$= \frac{\frac{2!}{(2-1)!*1!} * \frac{9!}{(9-2)!*2!}}{\frac{9!}{(9-3)!*3!}}$$

$$= \frac{7}{12}$$

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08-23
PNS

Q.6 → soln;

Total no. of members = 8

No. of civil engineer = 2

No. of IT officers = 4

No. of accountants = 2

No. of committee members to be chosen = 4

a) P(1 civil and 3 IT officers) = $\frac{{}^2C_1 * {}^4C_3}{{}^8C_4}$

$= \frac{\frac{2!}{(2-1)!*1!} * \frac{4!}{(4-3)!*3!}}{\frac{8!}{(8-4)!*4!}} = 4/35$

b) P(no civil) = $\frac{{}^6C_4}{{}^8C_4} = \frac{6!}{(6-4)!*4!} * \frac{(8-4)!*4!}{8!}$

$= 3/14$

c) P(atleast 1 IT officer) = $\frac{{}^4C_1 * {}^8C_3}{{}^8C_4}$

$= \frac{\frac{4!}{(4-1)!*1!} * \frac{8!}{(8-3)!*3!}}{\frac{8!}{(8-4)!*4!}}$

$= 69/70$

Ans

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08-23
PNS

(Q7) → soln;

Total no. of members = 12

No. of Nepalese = 3

No. of Indians = 4

No. of Americans = 5

No. of committee members to be chosen = 4

a) P(3 Nepalese and 1 Indian) = $\frac{{}^3C_3 * {}^4C_1}{{}^{12}C_4}$

$$= \frac{\frac{3!}{(3-3)! * 3!} * \frac{4!}{(4-1)! * 1!}}{\frac{12!}{(12-4)! * 4!}}$$

$$= \frac{4}{495}$$

b) P(1 Nepalese and 1 Indian and 2 American) = $\frac{{}^3C_1 * {}^4C_1 * {}^5C_2}{{}^{12}C_4}$

$$= \frac{\frac{3!}{(3-1)! * 1!} * \frac{4!}{(4-1)! * 1!} * \frac{5!}{(5-2)! * 2!}}{\frac{12!}{(12-4)! * 4!}}$$

$$= \frac{8}{33}$$

c) P(4 Americans) = $\frac{{}^5C_4}{{}^{12}C_4} = \frac{5!}{(5-4)! * 4!} * \frac{(12-4)! * 4!}{12!}$

$$= \frac{1}{99}$$

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* Laws of Addⁿ:

$P(A \cup B) = P(A) + P(B)$ [mutually exclusive]

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ [Not mutually exclusive]

* Multiplicative law:

$P(A \text{ and } B) = P(A \cap B) = P(A) \times P(B)$

(1) Ram and Hari appears for an interview for two posts. The probability of Ram selection is $\frac{1}{5}$ and Hari is $\frac{1}{7}$. Find the probability that

a) both of them selected → $P(A \cap B)$

b) Ram can select but Hari cannot → $P(A) \times \overline{P(B)}$

c) Hari can select but Ram cannot → $P(B) \times \overline{P(A)}$

d) None of them will be selected → $\overline{P(A)} \times \overline{P(B)}$

e) At least one of them will be selected → $P(A) \times \overline{P(B)} + P(B) \times \overline{P(A)} + P(A) \times P(B)$

f) only one of them will be selected.

↓

$P(A) \times \overline{P(B)} + P(B) \times \overline{P(A)}$

(2) The probability that a new airport will get an award for its design is 0.16, for materials is 0.21 and both awards is 0.11

a) What is the probability that it will get at least one of two awards? $P(d) \times \overline{P(m)} + P(\bar{d}) \times P(m) + P(d) \times P(m)$

b) What is the probability that it will get only one of two awards? $P(d) \times \overline{P(m)} + P(\bar{d}) \times P(m)$

(3) A problem in statistics is given to three students A, B, and C where chance of solving it are $\frac{1}{3}$, $\frac{1}{7}$, and $\frac{1}{5}$. Find the probability that

a) The problem will be solved. $P(A) \times P(B) \times P(C) \rightarrow$ ⊖

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08-24
PNS

b) Only one of them can solve the problem.
c) None of them can solve the problem.
d) A solve it but B and C can not.
e) All three can solve the problem.

$P(A) * P(\bar{B}) * P(\bar{C}) + P(\bar{A}) * P(B) * P(\bar{C}) + P(\bar{A}) * P(\bar{B}) * P(C)$

$P(A) * P(\bar{B}) * P(\bar{C})$

a) $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$

c) $\rightarrow P(\bar{A} \cap \bar{B} \cap \bar{C})$

*Conditional Probability:

$P(A/B) \rightarrow$ Probability of A Given B

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} \; ; \; P(B) \neq 0$$

$$P\left(\frac{B}{A}\right) = \frac{P(A \cap B)}{P(A)} \; ; \; P(A) \neq 0.$$

④ If $P(A) = \frac{3}{5}$, $P(B) = \frac{2}{5}$ and $P(A \text{ or } B) = \frac{4}{5}$. Find $P\left(\frac{A}{B}\right)$ and $P\left(\frac{B}{A}\right)$.

$P(A \text{ or } B) = P(A \cup B)$

⑤

Purchased DVD / Purchased TV	Yes (B)	No ($\bar{B}$)	total
Force TV (A)	38 ($A \cap B$)	42 ($A \cap \bar{B}$)	80
Not force TV ($\bar{A}$)	70 ($\bar{A} \cap B$)	150 ($\bar{A} \cap \bar{B}$)	220
total	108	192	300

Find:

a) probability that, if a household that purchased a television set is randomly selected, the television set purchased is force TV. $80/300$

b) find the probability that a randomly selected household that purchased a force TV and DVD players. $38/300$

c) what is probability that they purchased a force TV or DVD players? $80/300 + 108/300 - 38/300$

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d) What is the probability that a household purchased a DVD players given that household purchased a force TV? $\frac{38}{80}$

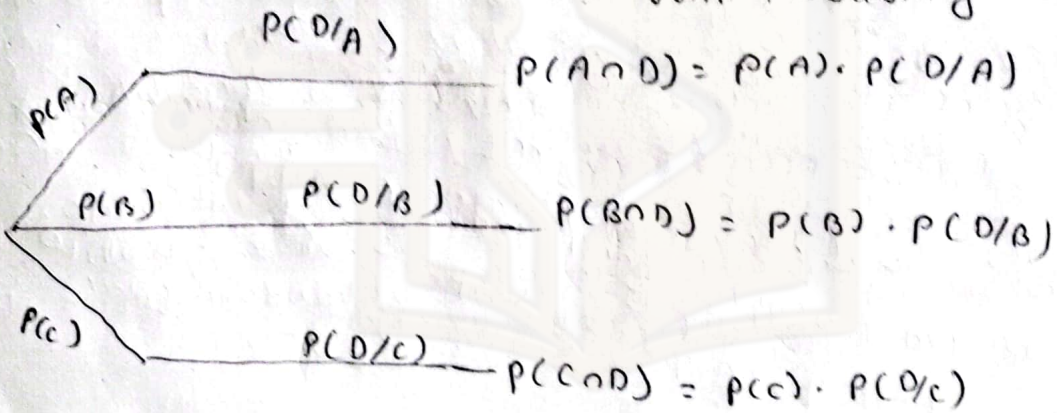
08-24
PN1

* Bayes's Theorem:

↳ If $E_1, E_2, E_3, \ldots E_n$ be mutually disjoint events of sample space S with $P(E_i) \neq 0$, $i = 1, 2, \ldots, n$ then for any arbitary events A which is subset of S such that $P(A) > 0$, we have

$$P(E_i/A) = \frac{P(E_i) \cdot P(A/E_i)}{\sum_{i=1}^{n} P(E_i) \cdot P(A/E_i)}$$

Joint Probability



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$$P(A/D) = \frac{P(A) \cdot P(D/A)}{P(A) \cdot P(D/A) + P(B) \cdot P(D/B) + P(C) \cdot P(D/C)}$$

$$P(B/D) = \frac{P(B) \cdot P(D/B)}{P(A) \cdot P(D/A) + P(B) \cdot P(D/B) + P(C) \cdot P(D/C)}$$

$$P(C/D) = \frac{P(C) \cdot P(D/C)}{P(A) \cdot P(D/A) + P(B) \cdot P(D/B) + P(C) \cdot P(D/C)}$$

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(Q.1) → Soln;

Probability of Ram selection P(R) = 1/5

Probability of Hari selection P(H) = 1/4

a) Probability of both of them selected is given by,

$$P(R \cap H) = P(R) * P(H)$$
$$= 1/5 * 1/4$$
$$= \frac{1}{20}$$

b) Probability of Ram select but not Hari is given by.

$$P_o(R) = P(R) * P(\bar{H})$$
$$= \frac{1}{5} * \left(1 - \frac{1}{4}\right)$$
$$= \frac{3}{20}$$

c) Probability of Hari select but not Ram is given by

$$P_o(H) = P(H) * P(\bar{R})$$
$$= \frac{1}{4} * \left(1 - \frac{1}{5}\right)$$
$$= \frac{4}{20}$$

d) Probability of none selection is given by.

$$P(\bar{R} \cap \bar{H}) = P(\bar{R}) * P(\bar{H})$$
$$= \left(1 - \frac{1}{5}\right) * \left(1 - \frac{1}{4}\right)$$
$$= \frac{3}{5}$$

e) Probability of at least one selection is given by.

$$P_o(R) + P_o(H) + P(R \cap H)$$
$$= \frac{3}{20} + \frac{4}{20} + \frac{1}{20}$$
$$= 2/5$$

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f) Probability of only one selection is given by.

$$P_o(A) + P_o(H)$$

$$= \frac{3}{20} + \frac{1}{20}$$

$$= 7/20 \quad \text{Ans}$$

(Q.2) → soln;

Probability of getting award for design $P(D) = 0.16$

Probability of getting award for material $P(M) = 0.24$

Probability of getting award for both $P(D \cap M) = 0.11$

a) Probability of getting at least one award is given by.

$$P_o(D) + P_o(M) + P(D \cap M)$$

$$= 0.16 * (1 - 0.24) + 0.24 * (1 - 0.16) + 0.11$$

$$= 0.4332$$

b) Probability of getting only one award is given by

$$P_o(D) + P_o(M)$$

$$= 0.16 * (1 - 0.24) + 0.24 * (1 - 0.16)$$

$$= 0.3232 \quad \text{Ans}$$

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(Q.3) → soln;

Probability of solving by A, $P(A) = 1/3$

Probability of solving by B, $P(B) = 1/4$

Probability of solving by C, $P(C) = 1/5$

a) Probability of problem being solved is given by.

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$$

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$= \frac{1}{3} + \frac{1}{4} + \frac{1}{5} - \left(\frac{1}{3} * \frac{1}{4}\right) - \left(\frac{1}{4} * \frac{1}{5}\right) - \left(\frac{1}{3} * \frac{1}{5}\right)$ 08-24 PNS

$+ \left(\frac{1}{3} * \frac{1}{4} * \frac{1}{5}\right)$

$= 3/5$

b) Probability of only one ~~selection~~ can solve is given by

~~P of A n B n~~ $P_o(A) + P_o(B) + P_o(C)$

$= P(A) * P(\bar{B}) * P(\bar{C}) + P(\bar{A}) * P(B) * P(\bar{C}) + P(\bar{A}) * P(\bar{B}) * P(C)$

$= \frac{1}{3} * \left(1 - \frac{1}{4}\right) * \left(1 - \frac{1}{5}\right) + \left(1 - \frac{1}{3}\right) * \frac{1}{4} * \left(1 - \frac{1}{5}\right) + \left(1 - \frac{1}{3}\right) * \left(1 - \frac{1}{4}\right) * \frac{1}{5}$

$= 13/30$

c) Probability of none can solve is given by.

$P(\bar{A} \cap \bar{B} \cap \bar{C}) = P(\bar{A}) * P(\bar{B}) * P(\bar{C})$

$= \left(1 - \frac{1}{3}\right) * \left(1 - \frac{1}{4}\right) * \left(1 - \frac{1}{5}\right)$

$= 2/5$

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d) Probability of solved by A but not by B and C is given by

$P_o(A) = P(A) * P(\bar{B}) * P(\bar{C})$

$= \frac{1}{3} * \left(1 - \frac{1}{4}\right) * \left(1 - \frac{1}{5}\right)$

$= 1/5$

e) Probability of All three can solve is given by.

$P(A \cap B \cap C) = P(A) * P(B) * P(C)$

$= \frac{1}{3} * \frac{1}{4} * \frac{1}{5} = \frac{1}{60}$ Ans

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08-24
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(Q.3) → soln;

a) ~~Probability~~ No. of favorable cases = 80

Total cases = 300

Probability of purchased television to be force TV is given by

$$P(F) = \frac{80}{300}$$

$$= \frac{4}{15}$$

b) No. of favourable cases = 38

Total cases = 300

Probability of purchasing force TV with DVD is given by

$$P(F \cap D) \quad \frac{38}{300}$$

$$= \frac{19}{150}$$

c) Probability of purchasing force tv or DVD is given by

$$P(F) + P(D) - P(F \cap D)$$

$$= \frac{4}{15} + \frac{19}{150} - \frac{108}{300} - \frac{19}{150}$$

$$= \frac{1}{2}$$

d) Probability of purchasing DVD given that household purchased a force TV is given by.

$$P(D/F) = \frac{P(F \cap D)}{P(F)}$$

$$= \frac{19}{150} * \frac{15}{4}$$

$$= \frac{19}{40} \quad \text{Ans}$$

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Imp

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08-25
PNS

① In a bolt factory machine A, B, C manufacture respectively 25%, 35% and 40% of the total of their output. 5, 4, 2% are defective bolts. A bolt is drawn at random from the product and is found to be defective. What is the probability that it was manufactured by machine A?

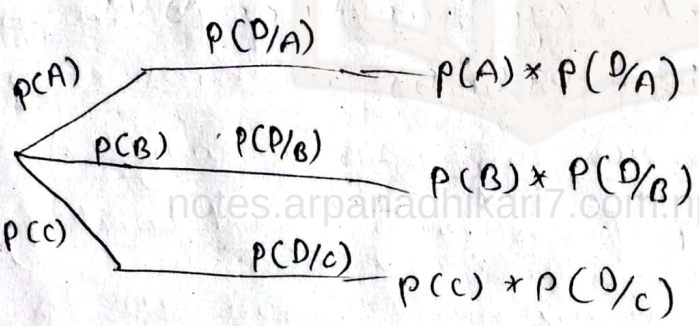
Imp

② The chances of x, y, z becoming manager of a certain company are 1:2:3. The probability that bonus scheme will be introduced if x, y, z becomes manager are 0.3, 0.5 and 0.8 respectively. If the bonus scheme has been introduced, what is the probability that x is appointed as the manager?

① → Soln;

$P(A) = 0.25$, $P(B) = 0.35$, $P(C) = 0.40$

$P(D/A) = 0.05$ $P(D/B) = 0.04$ $P(D/C) = 0.02$



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Now,

$$P(A/D) = \frac{P(A) * P(D/A)}{P(A)*P(D/A) + P(B)*P(D/B) + P(C)*P(D/C)}$$

$$= \frac{0.25 * 0.05}{0.25*0.05 + 0.35*0.04 + 0.40*0.02}$$

$$= \frac{25}{69}$$ Ans

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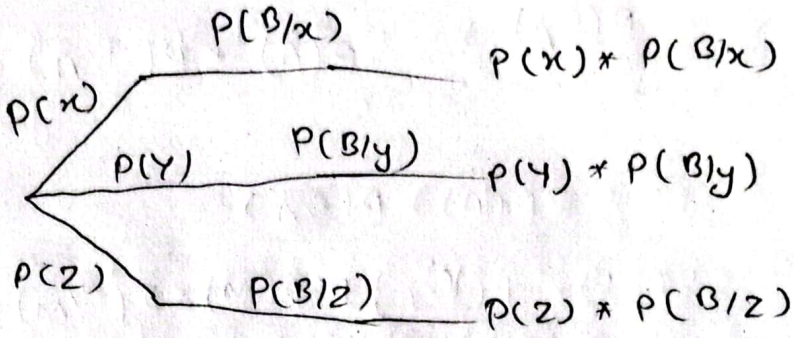
08-25
PNS

(2) → soln;

$P(X) = \frac{4}{9}$, $P(Y) = \frac{2}{9}$, $P(Z) = \frac{3}{9}$

$P(B/X) = 0.3$, $P(B/Y) = 0.5$, $P(B/Z) = 0.8$



Now,

$$P(X/B) = \frac{P(x) * P(B/x)}{P(x) * P(B/x) + P(Y) * P(B/Y) + P(Z) * P(B/Z)}$$

$$= \frac{4/9 * 0.3}{4/9 * 0.3 + 2/9 * 0.5 + 3/9 * 0.8}$$

$$= 6/23$$ Ans

(3) In a class of 75 students, 15 students were considered to be very intelligent, 45 medium and rest below the average. The probability that a very intelligent student fail in exam is 0.005, the medium student fail is 0.05, and corresponding probability for below average is 0.15. If a student is known to have pass the exam, what is the probability that he is below the average.

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→ soln;

$P(I) = \frac{15}{75}$ $P(M) = \frac{45}{75}$ $P(A) = \frac{15}{75}$

$P(F/I) = 0.005$ $P(F/M) = 0.05$ $P(F/A) = 0.15$

$P(P/I) = 0.995$ $P(P/M) = 0.95$ $P(P/A) = 0.85$

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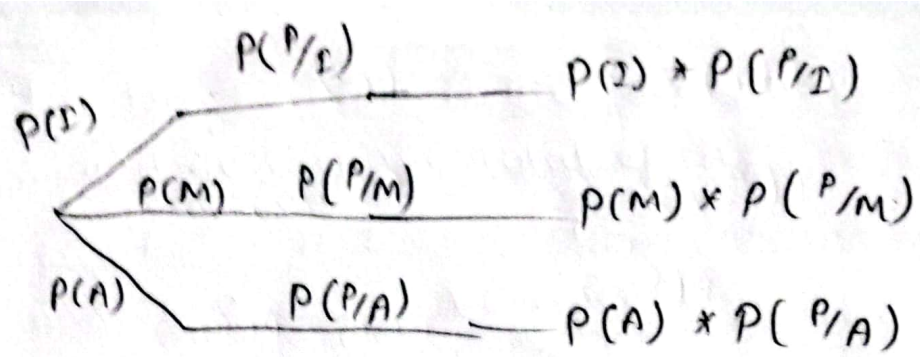

Now,

$$P(A/P) = \frac{P(A) * P(P/A)}{P(I) * P(P/I) + P(M) * P(P/M) + P(A) * P(P/A)}$$

$$= \frac{\frac{15}{75} * 0.85}{\frac{15}{75} * 0.995 + \frac{45}{75} * 0.95 + \frac{15}{75} * 0.85}$$

$$= \frac{170}{939} = 0.181$$

Ans 2

IMP

④ In a certain requirement test, there are multiple choice question. There are 4 possible answer to each question and of which one is correct. An intelligent knows 80% of answer. If intelligent student get correct answer, what is probability that he is guessing? A weak student knows 10% of answer. If a weak student gets correct answer what is the probability that he is guessing?

Case-I:

$P(I) = 0.8$ $\quad P(G) = 0.2$

$P(E_1/C) = 1$ $\quad P(E_2/G) = 1/4$

$$P(G/E_2) = \frac{P(G) \cdot P(E_2/G)}{P(C) * P(E_1/C) + P(G) * P(E_2/G)}$$

$$= \frac{0.2 * 1/4}{0.8 * 1 + 0.2 * 1/4} = 1/17 = 0.058$$

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Case-II

08-25
PNS

$P(C) = 0.1$ $\quad P(\bar{C}) = 0.9$

$P(E_2/C) = 1$ $\quad P(E_2/\bar{C}) = 1/4$

$$P(\bar{C}/E_2) = \frac{P(\bar{C}) * P(E_2/\bar{C})}{P(C) * P(E_2/C) + P(\bar{C}) * P(E_2/\bar{C})}$$

$$= \frac{0.9 * 1/4}{0.1 * 1 + 0.9 * 1/4} = \frac{9}{13} = 0.692$$ Ans

* Expected value of discrete random variable:

Definition:

If X is a discrete random variable assumes the values $x_1, x_2, \ldots$ along with corresponding probabilities $P(x_1)$, $P(x_2)$, ... then the expection or mean of X denoted by $E(x)$ is defined by

$$\mu = E(x) = \sum x_i * P(x_i)$$

It is also called mathematical expectation or mean of X.

Properties of Expected values:

1. $E(X+Y) = E(X) + E(Y)$
2. $E(XY) = E(X) * E(Y)$
3. $\sigma^2 = E(x^2) - [E(x)]^2$
4. $E(ax+b) = a \cdot E(x) + b$
5. $V(ax+b) = a^2 \cdot V(x)$, $[b = 0]$

(1) A random variable x has following proof Imp

$X = x$	-2	-1	0	1	2	3
$P(X = x)$	0.1	k	0.2	2k	0.3	k

Ckl

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Find: i) The value of k

ii) $P(-1 \leq x \leq 2)$

iii) mean and variance

iv) $E(4x+5)$

v) $V(4x+5)$

↳ soln;

i) we have,

total probability function = 1

or, $\Sigma P(x_i) = 1$

or, $0.1 + k + 0.2 + 2k + 0.3 + k = 1$

or, $0.6 + 4k = 1$

$\therefore k = 0.1$

Calculation table

$X = x$	$P(X=x)$	x^2	$x \cdot P(x)$	$x^2 \cdot P(x)$
-2	0.1	4	-0.2	0.4
-1	0.1	1	-0.1	0.1
0	0.2	0	0	0
1	0.2	1	0.2	0.2
2	0.3	4	0.6	1.2
3	0.1	9	0.3	0.9
	1		$\Sigma x_i \cdot P(x_i) = 0.8$	$\Sigma x_i^2 \cdot P(x_i) = 2.8$

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ii) $P(-1 \leq x \leq 2) = 0.1 + 0.2 + 0.2 + 0.3$

$= 0.8$

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08-25
PNS

iii) Mean : $E(x) = \sum x_i \cdot p(x_i)$

$= 0.8$

Variance $E(x^2) = \sum x_i^2 \cdot p(x_i)$

$= 2.8$

Variance $V(x) = E(x^2) - [E(x)]^2$

$= 2.8 - (0.8)^2$

$= 2.16$

iv) $E(4x+5) = 4 \cdot E(x) + 5$

$= 4 * 0.8 + 5$

$= 8.2$

v) $V(4x+5) = 4^2 * V(x)$

$= 16 * 2.16$

$= 34.56$

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